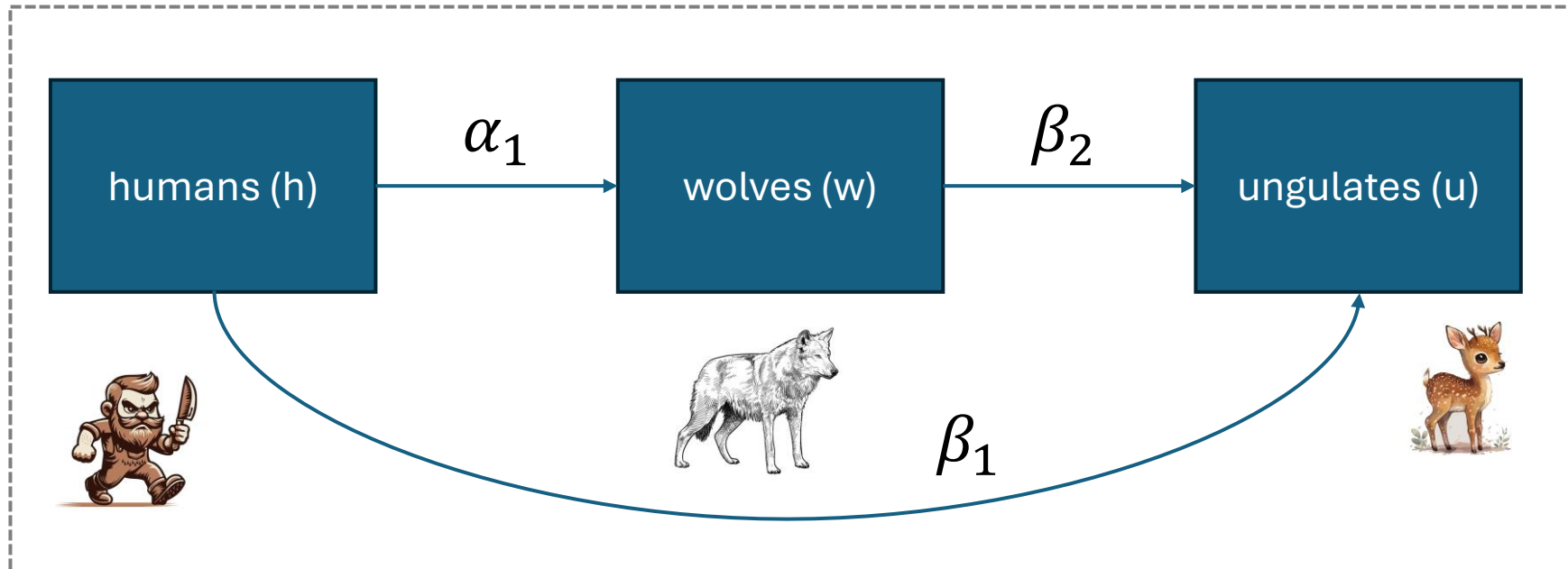
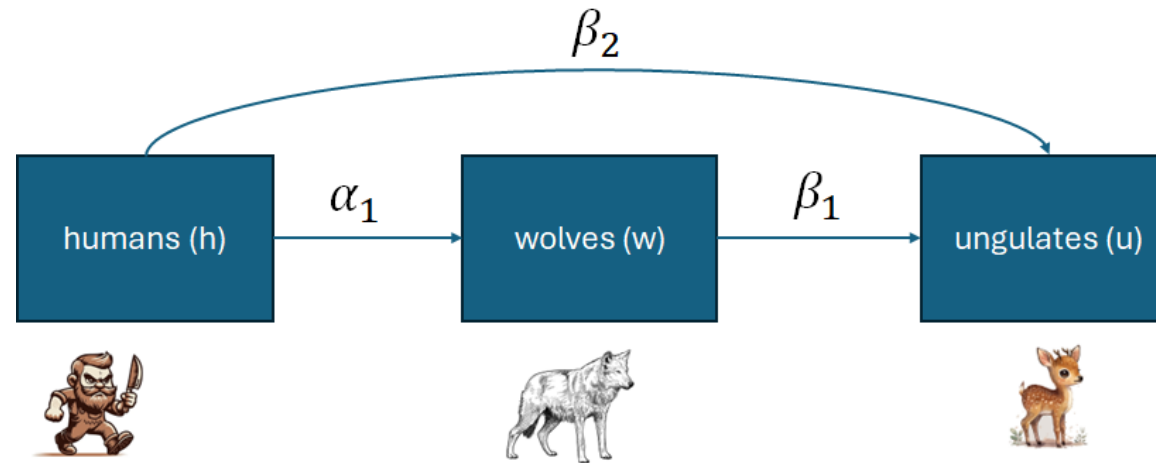


Direct and indirect effects (WILD 595)



Let's simulate some data ($n = 100$) and talk about it.

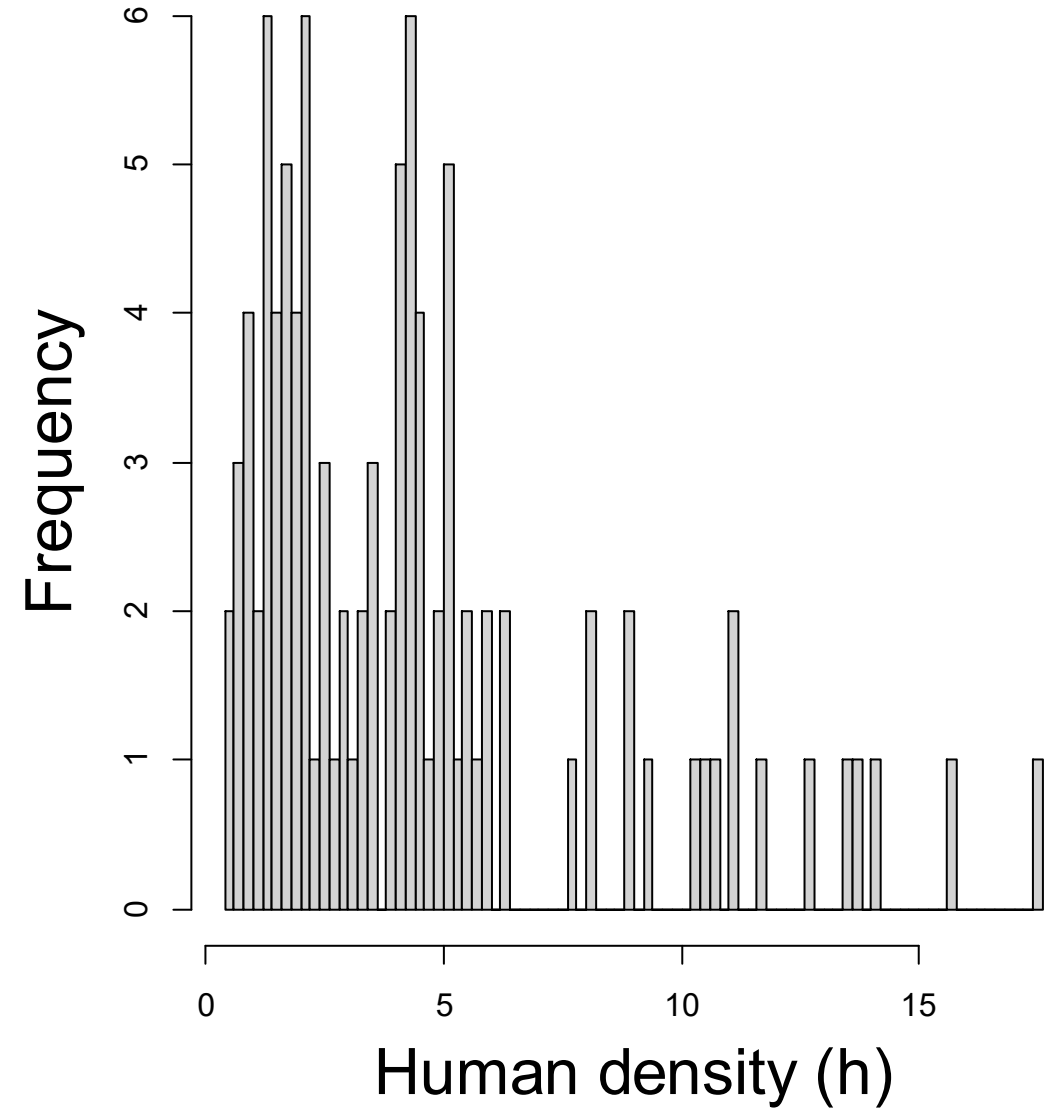
We'll simulate human (h), wolf (w), and ungulate (u) abundance along a 'wild-to-suburban' gradient



Simulating the data: humans

$$h \sim \text{lognormal}(1, 0.75)$$

humans (h)



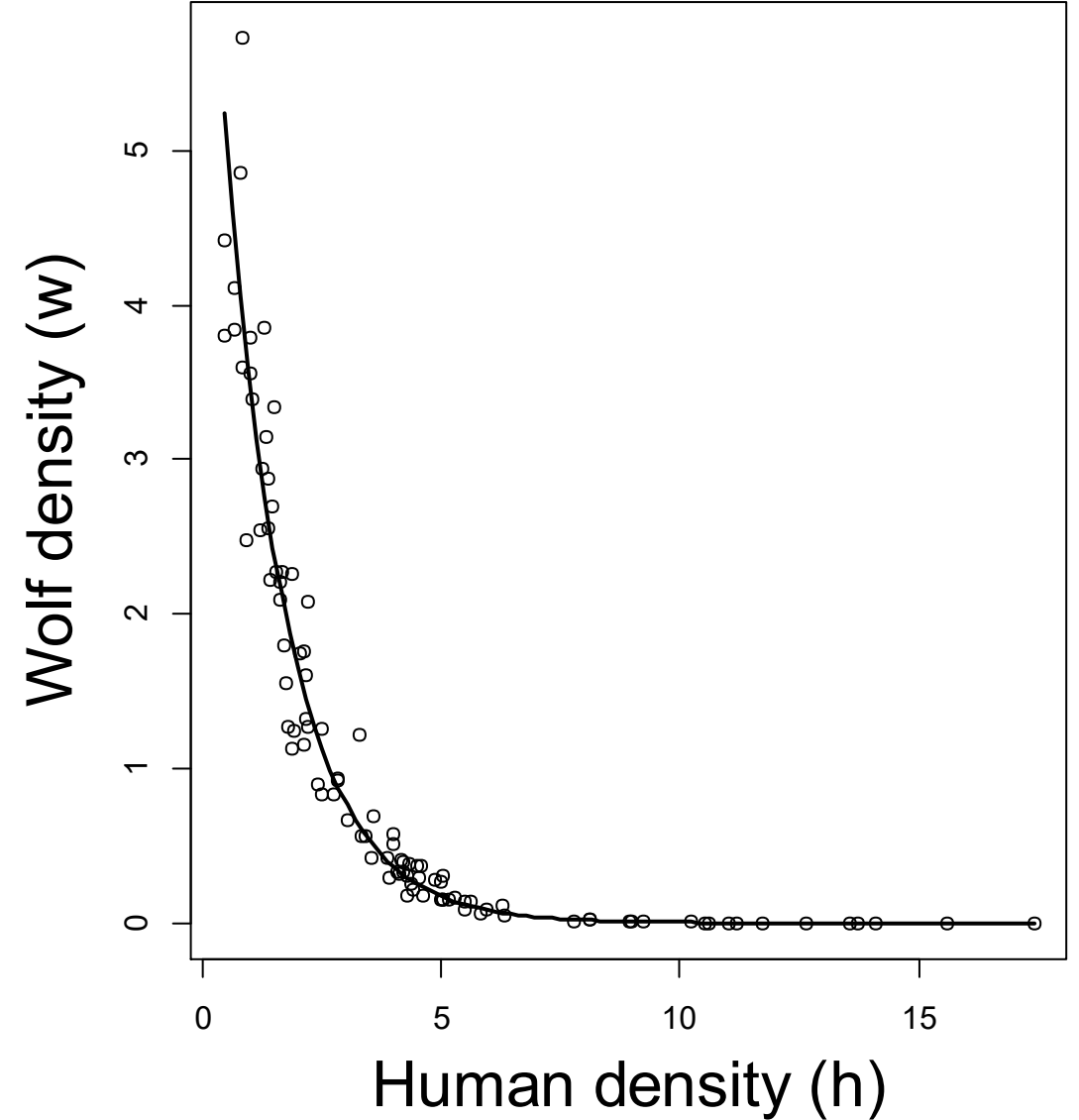
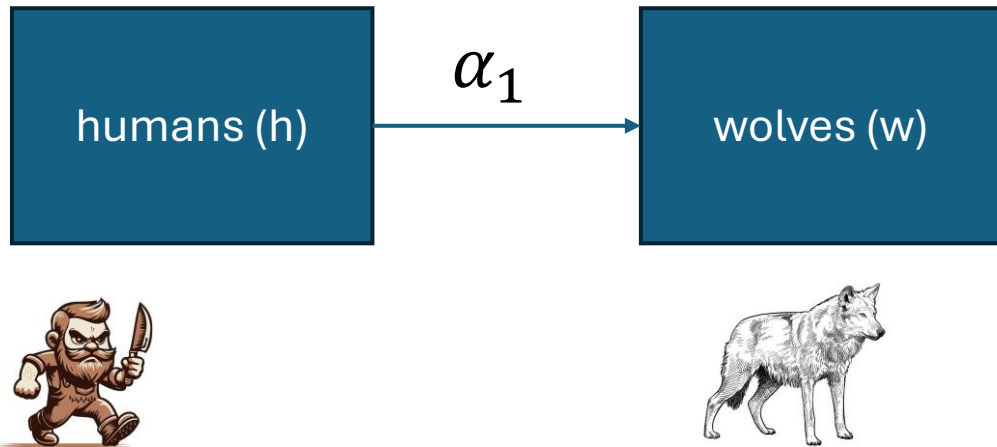
We'll assume these sites have similar habitat types/qualities

Simulating the data: wolves

$$w \sim \text{lognormal}(\alpha_0 + \alpha_1 \mathbf{h}, \sigma_w^2)$$

$$\alpha_0 = 3$$

$$\alpha_1 = -0.75$$



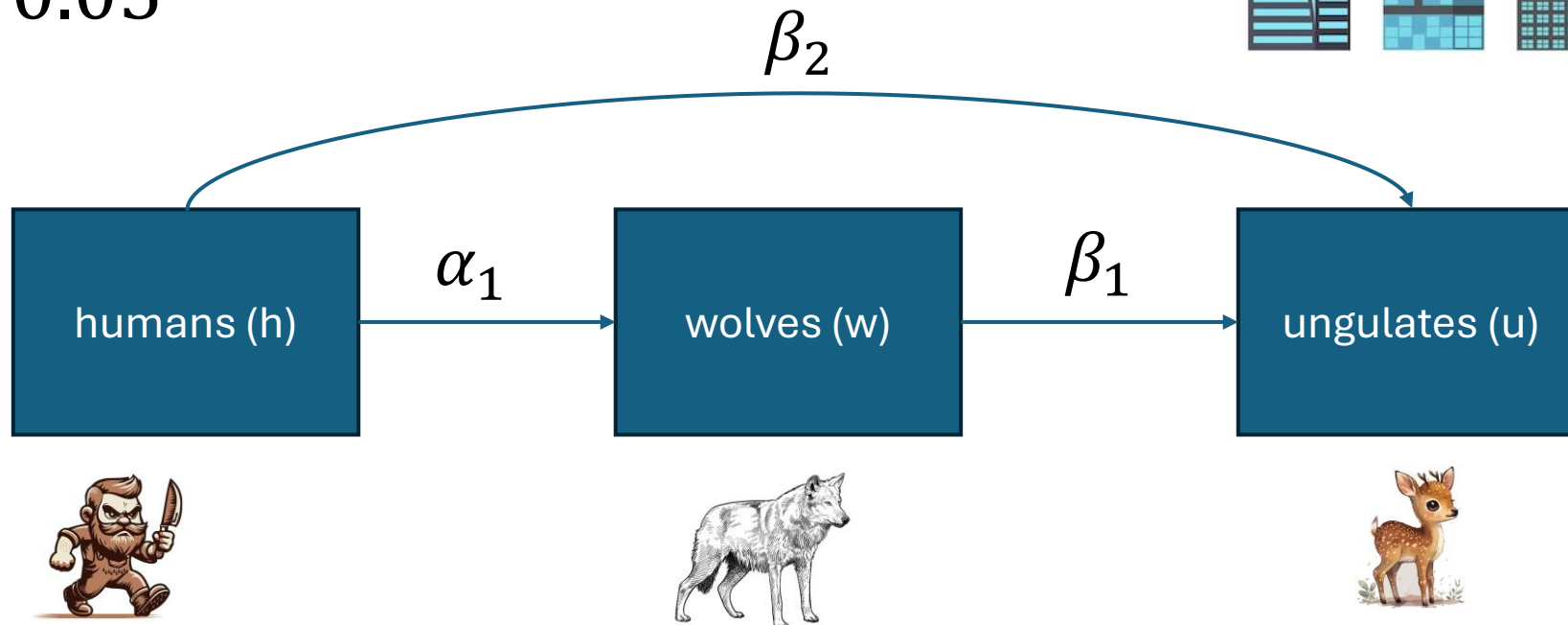
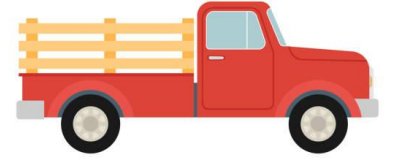
Simulating the data: ungulates

$$u \sim \text{lognormal}(\beta_0 + \beta_1 \mathbf{w} + \beta_2 \mathbf{h}, \sigma_u^2)$$

$$\beta_0 = 3$$

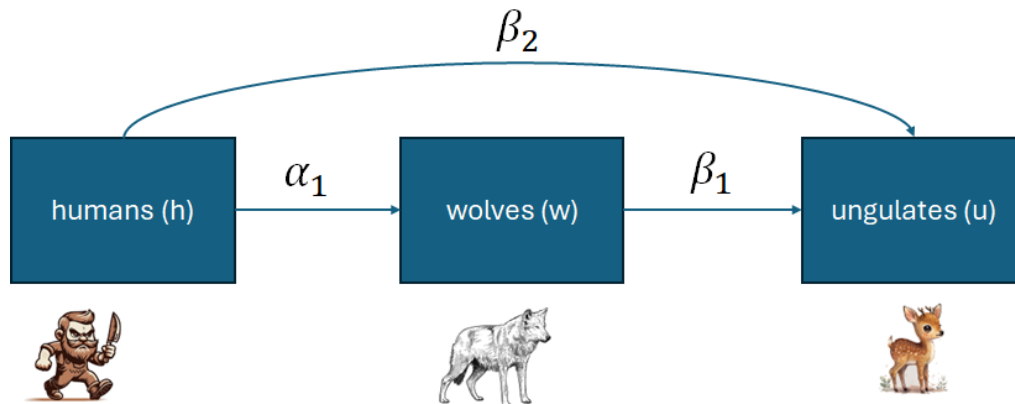
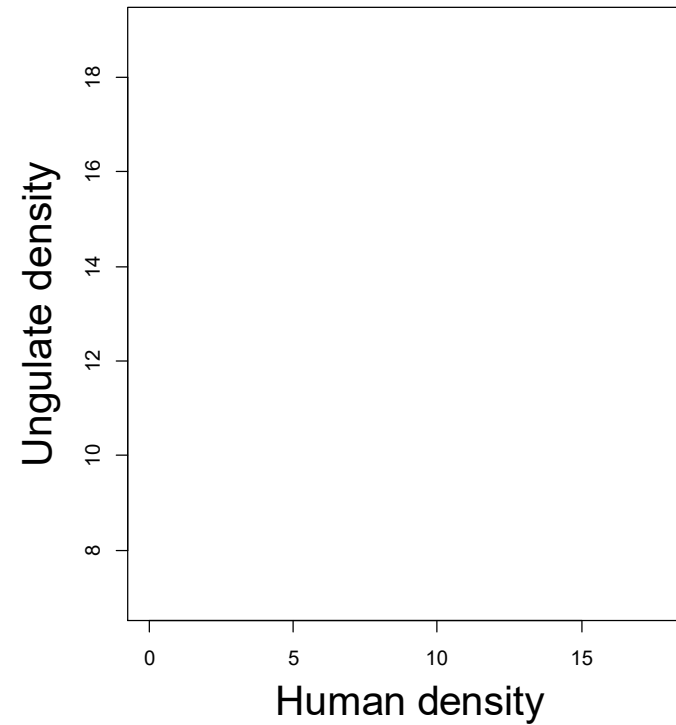
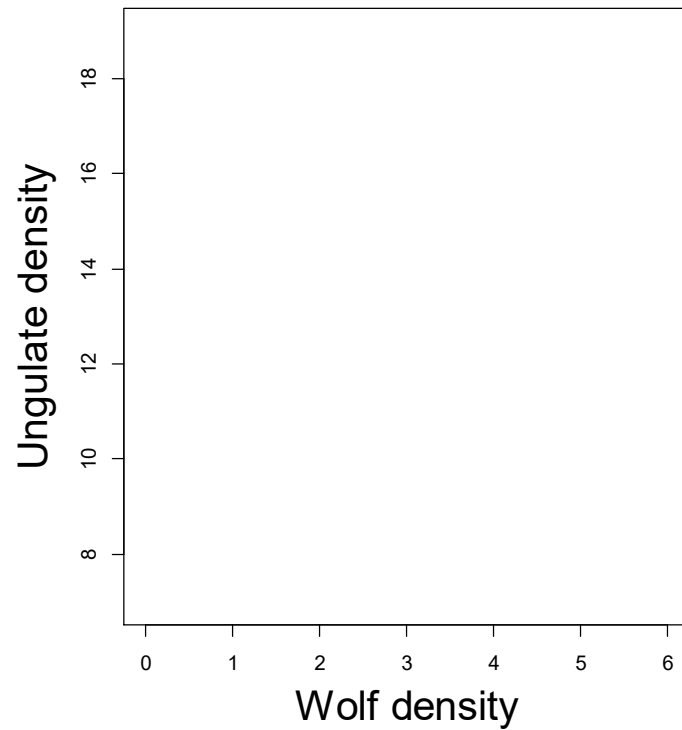
$$\beta_1 = -0.15$$

$$\beta_2 = -0.05$$



Now let's look at our observed relationships

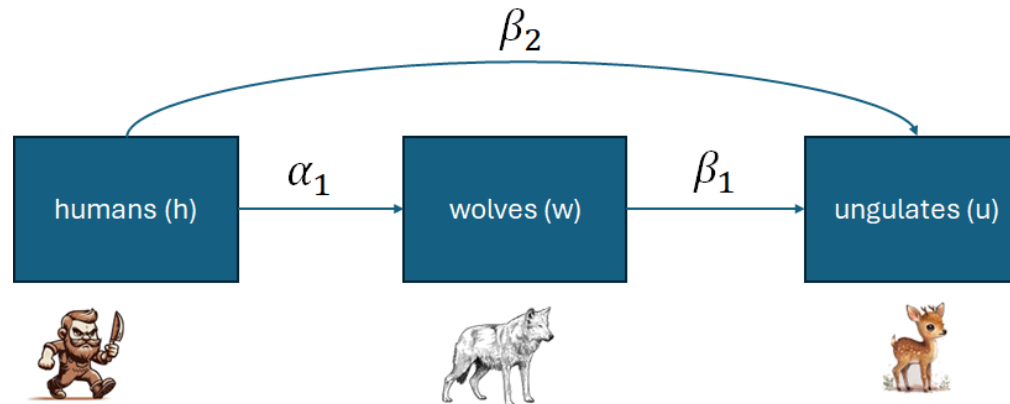
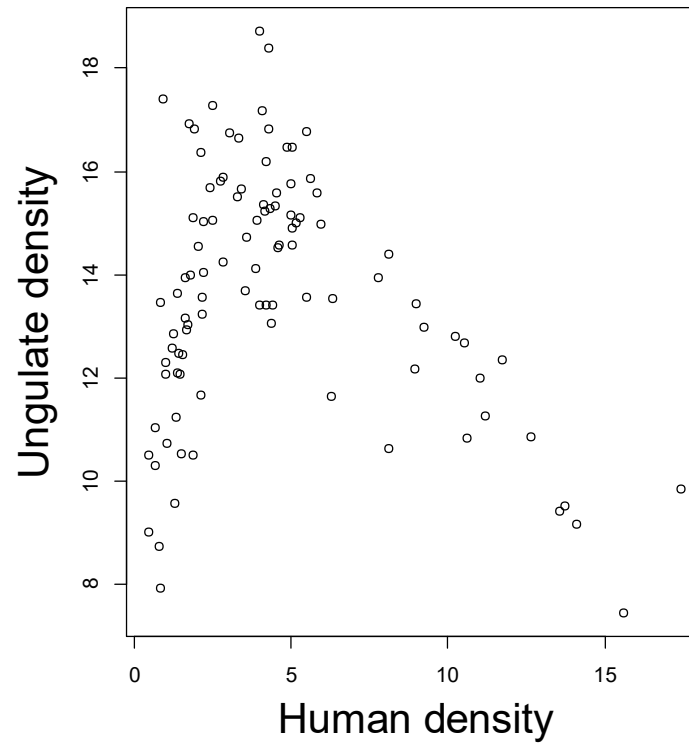
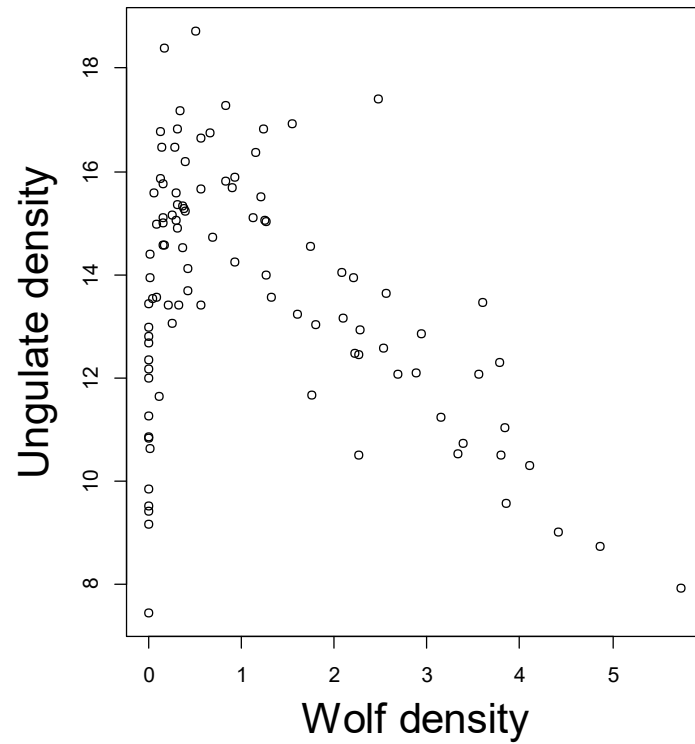
What do we expect to see?



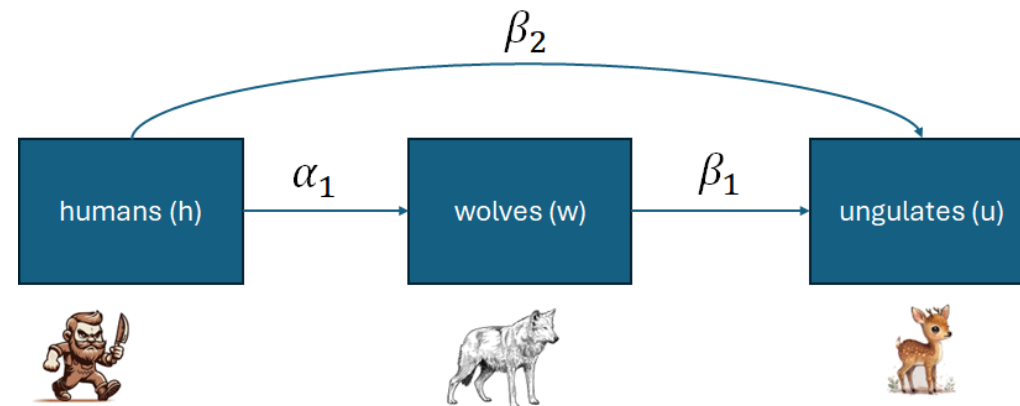
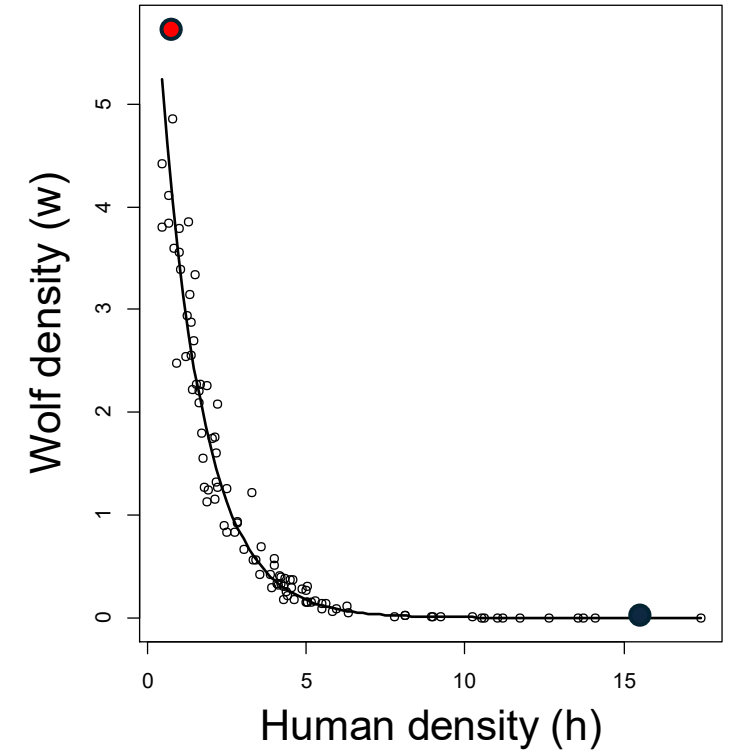
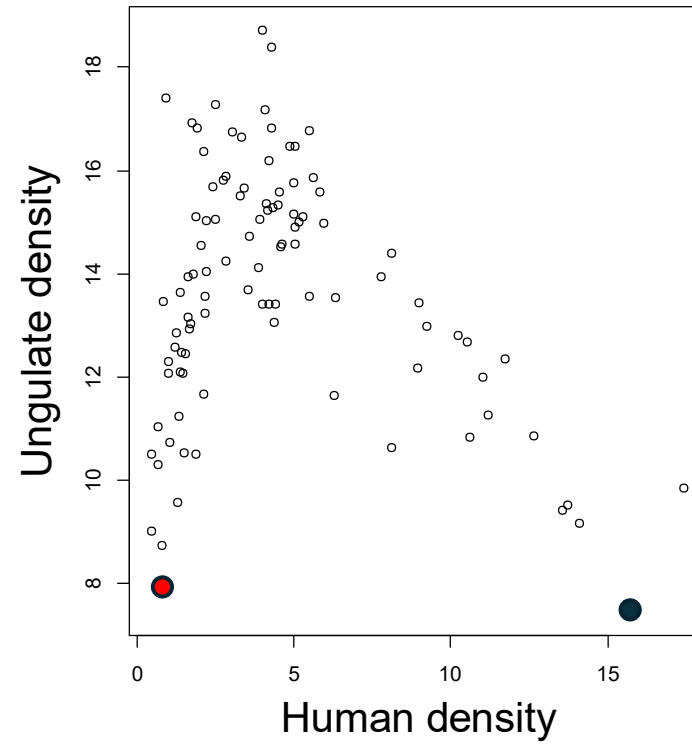
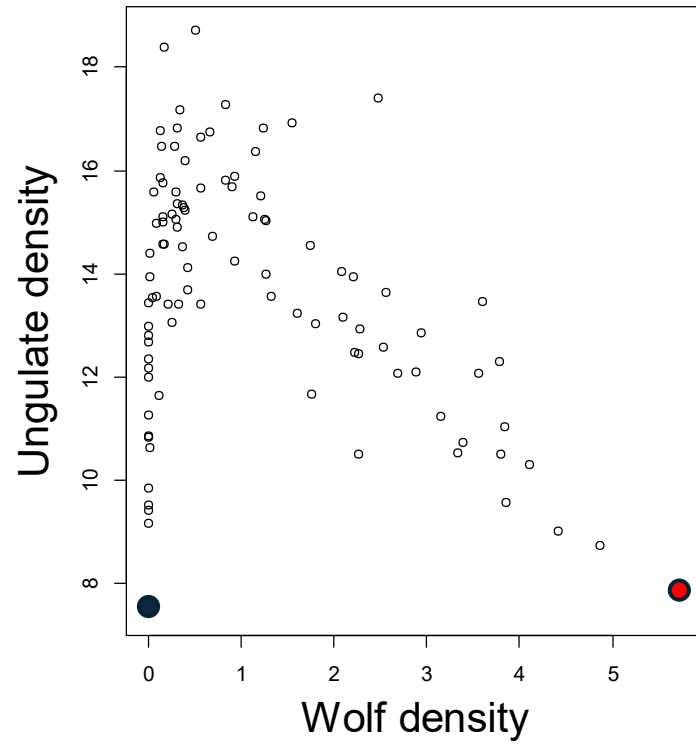
$$\beta_1 = -0.15$$

$$\beta_2 = -0.05$$

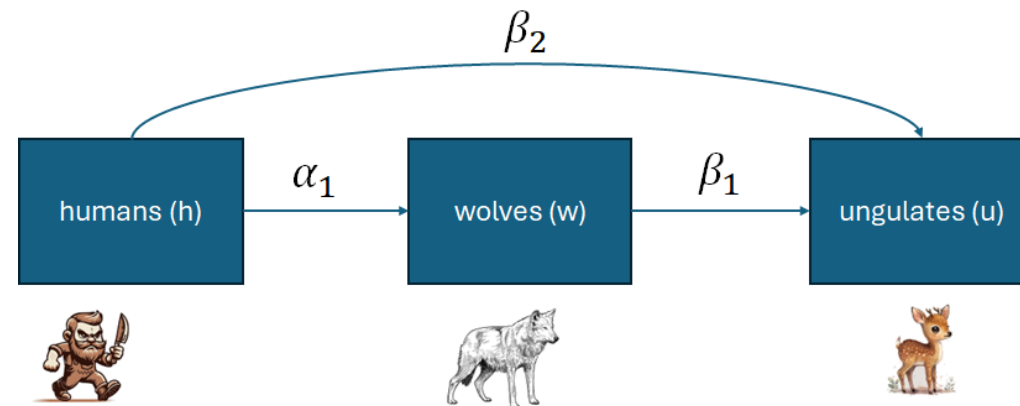
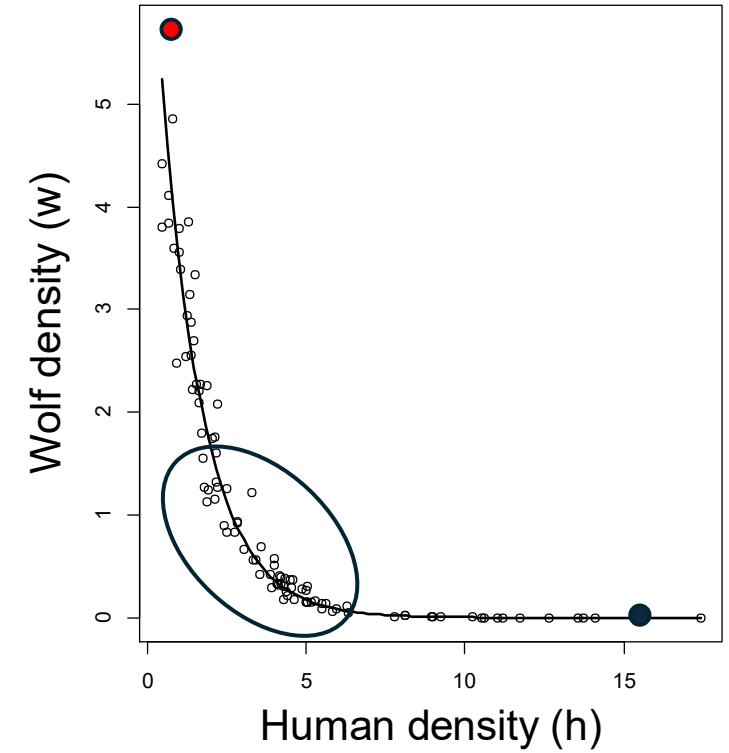
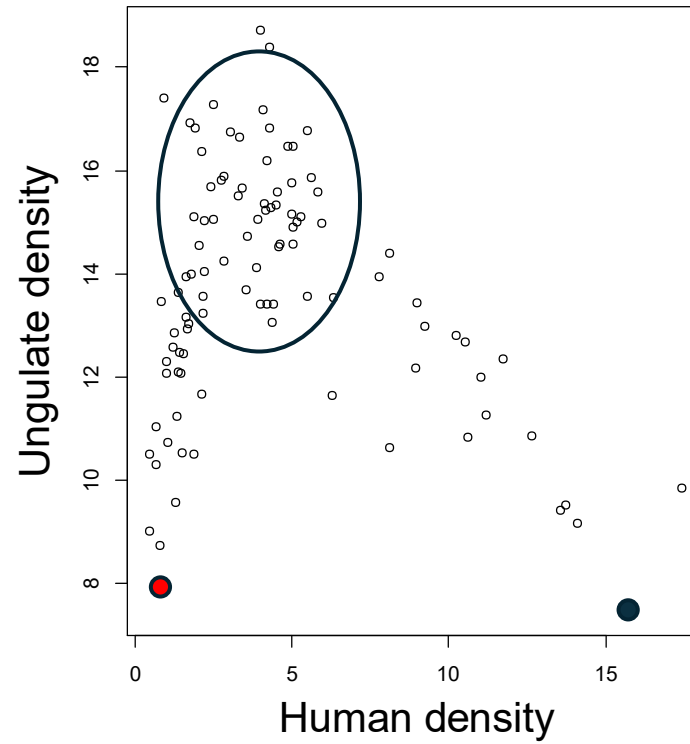
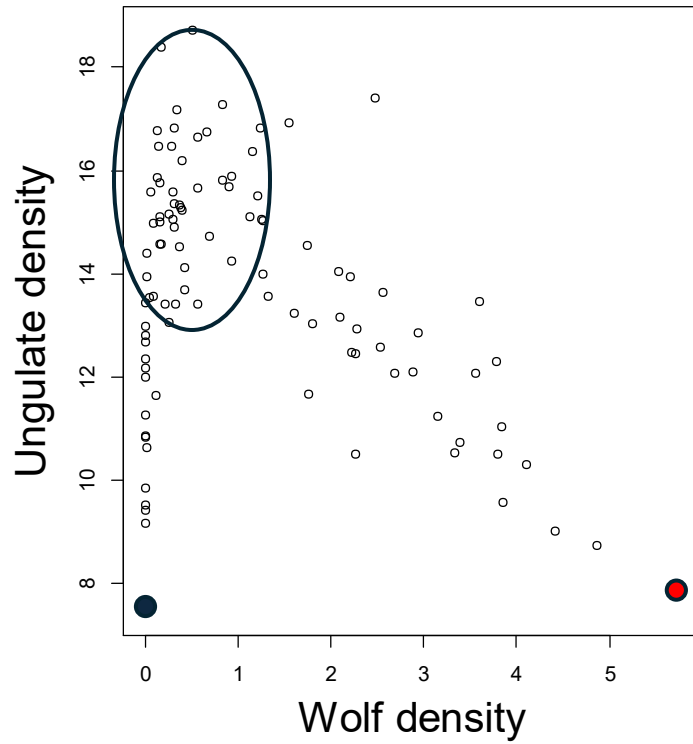
What's happening here?!



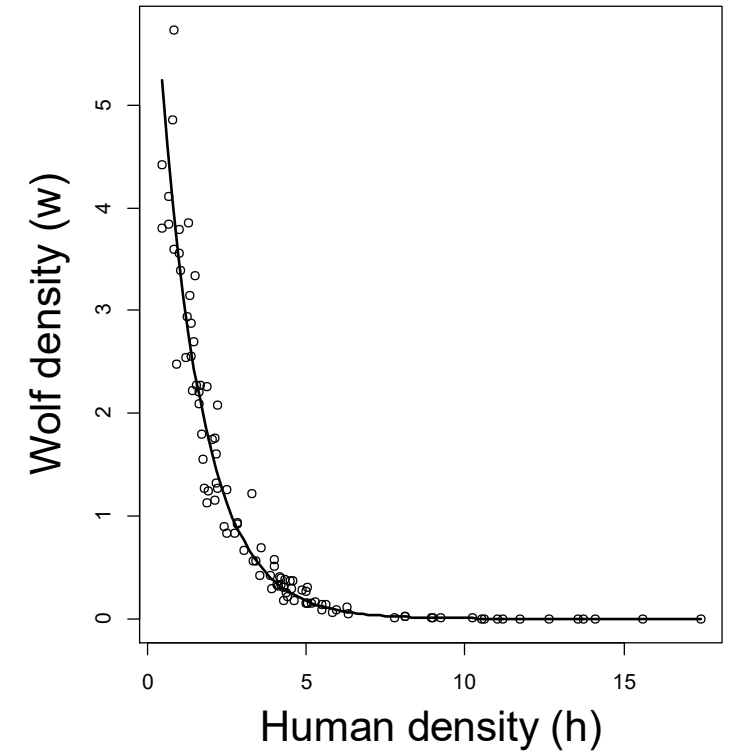
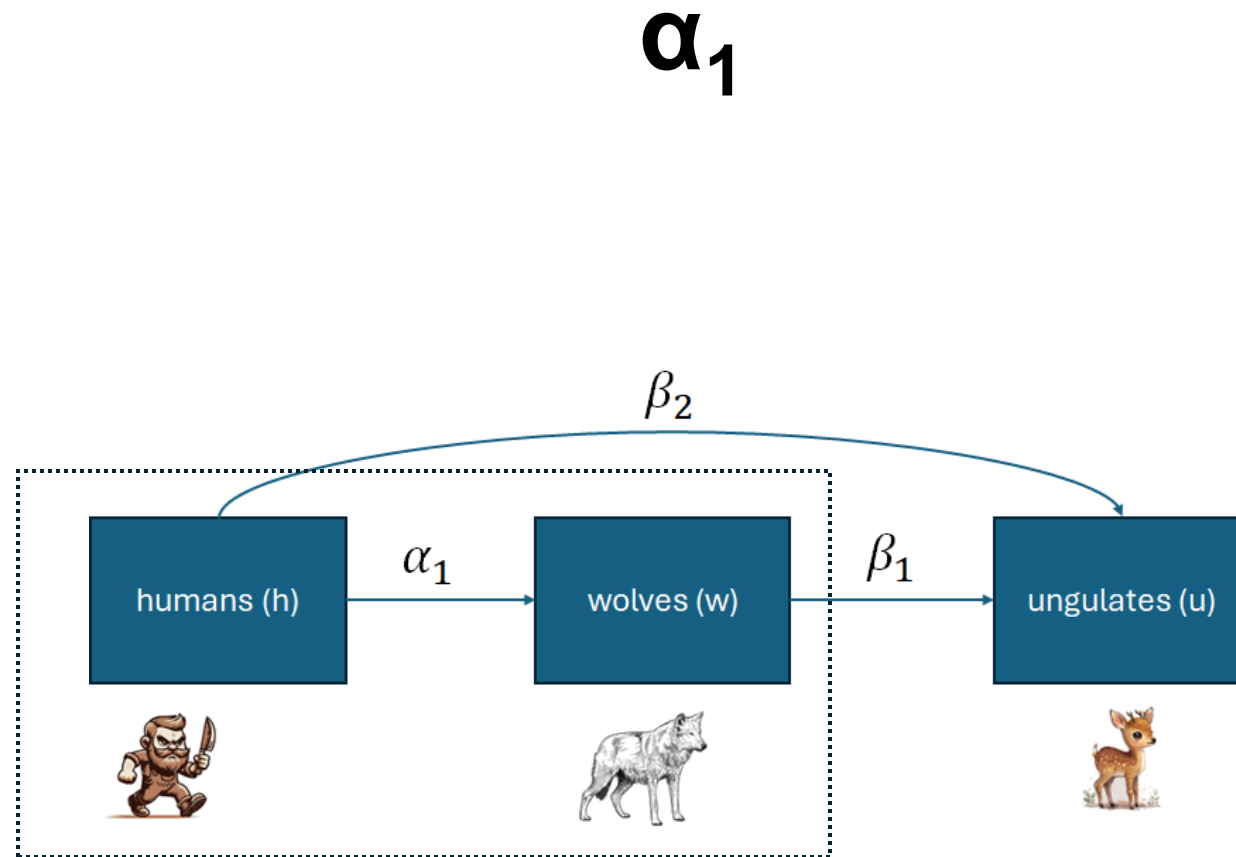
What's happening here?!



What's happening here?!

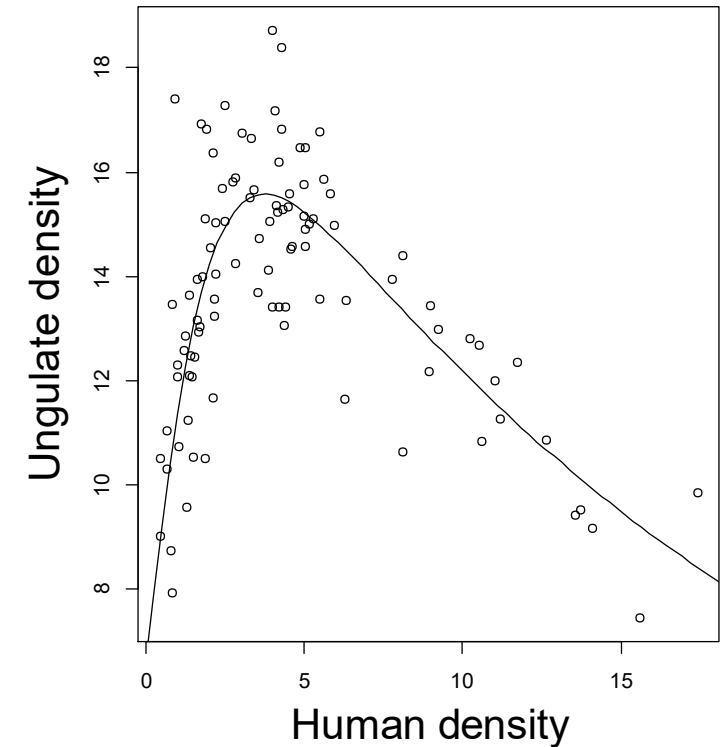
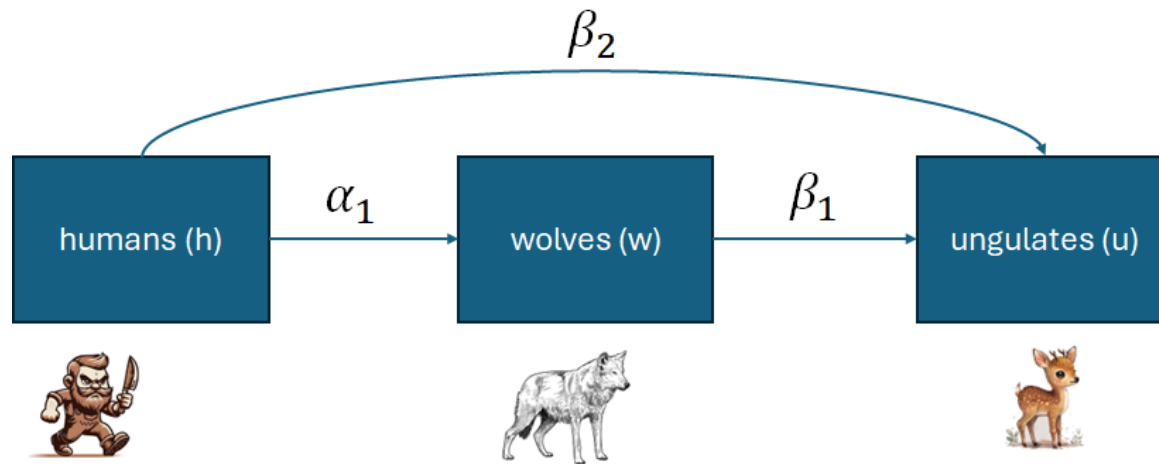


The total effect of humans on wolves is easy to estimate



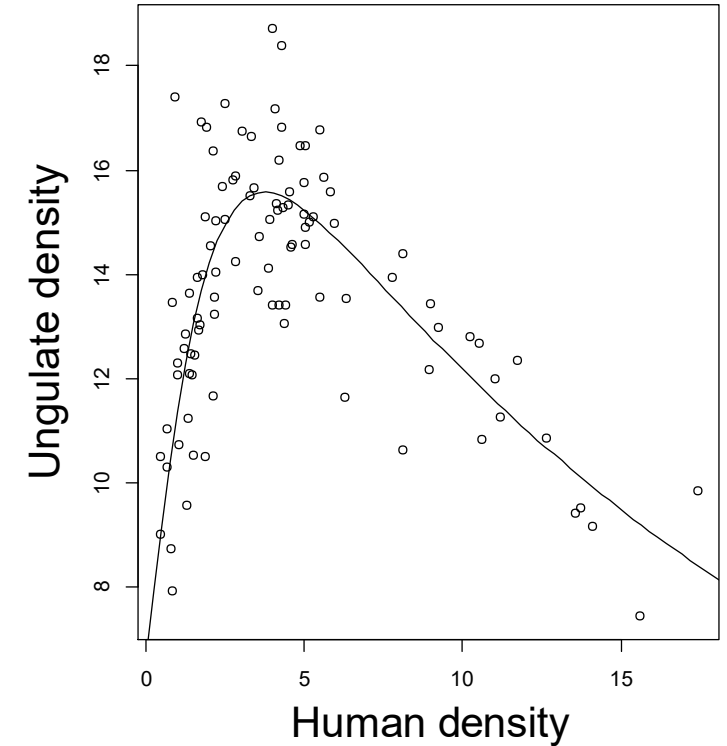
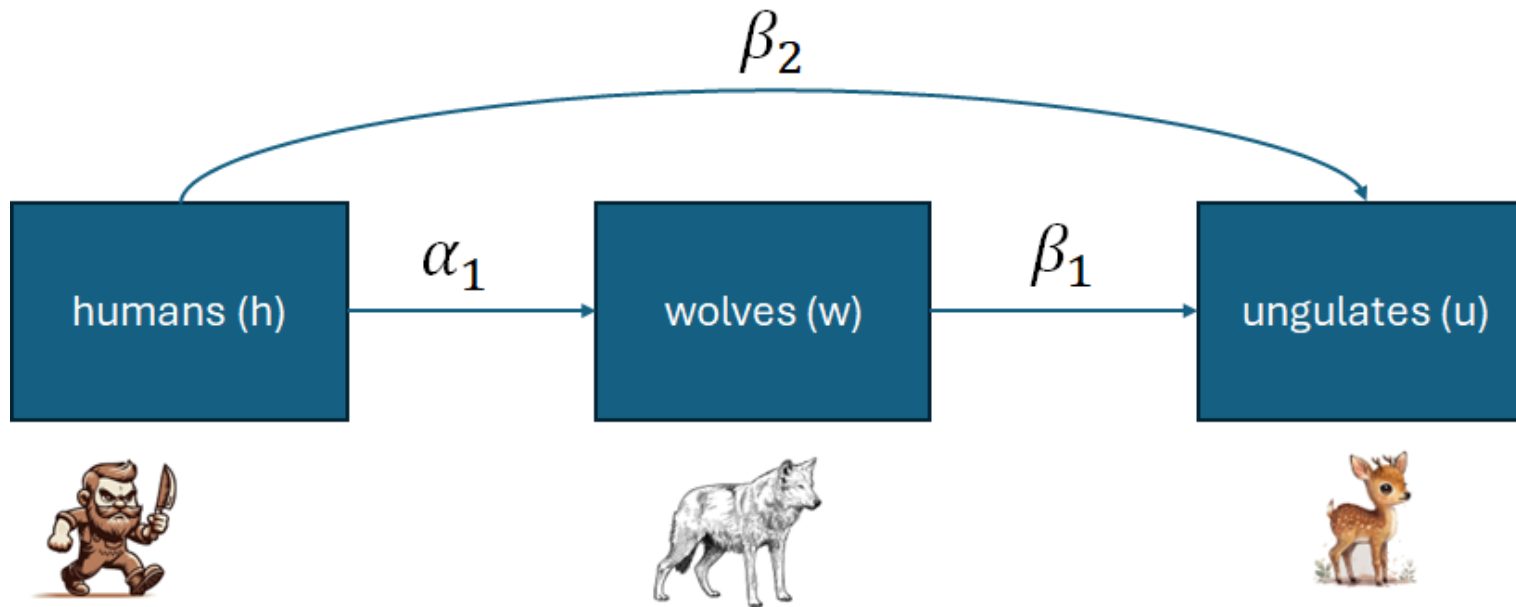
The total effect of humans on ungulates is...

A function of both β_2 (the direct effect) as well as the indirect pathway through wolves (a 'mediator') which is a function of α_1 and β_1



If identity link, simply $\alpha_1\beta_1 + \beta_2$ (we won't use that link much...)

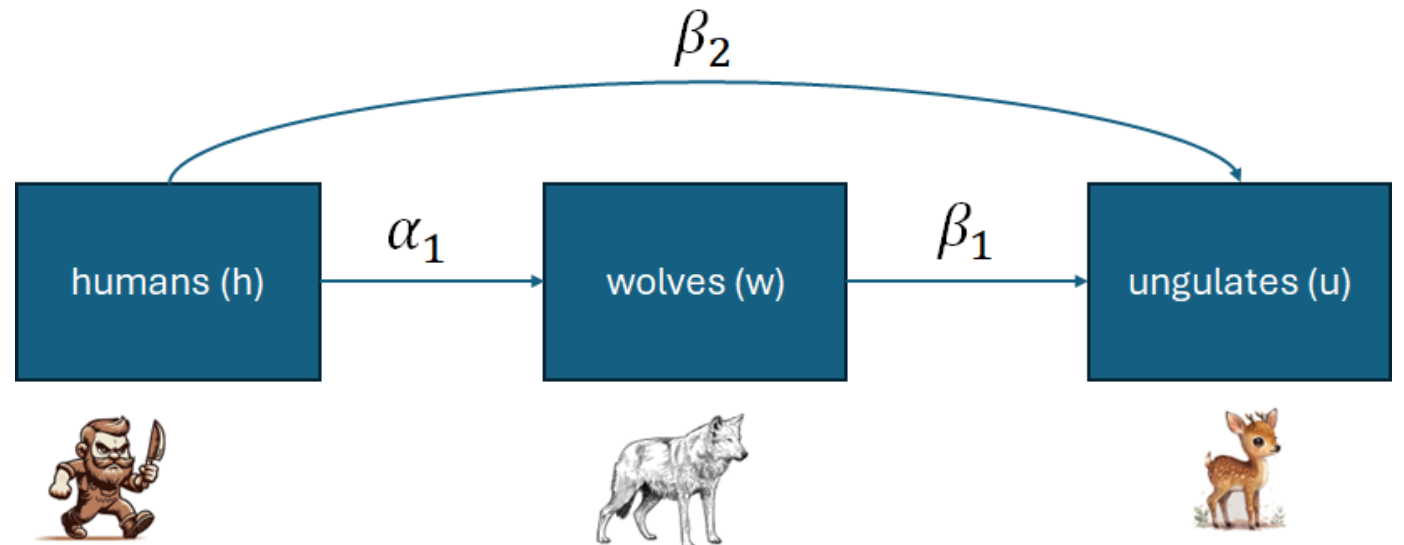
Direct and indirect effects



α_1 , β_1 , & β_2 are direct effects, the indirect effect is a function of α_1 & β_1

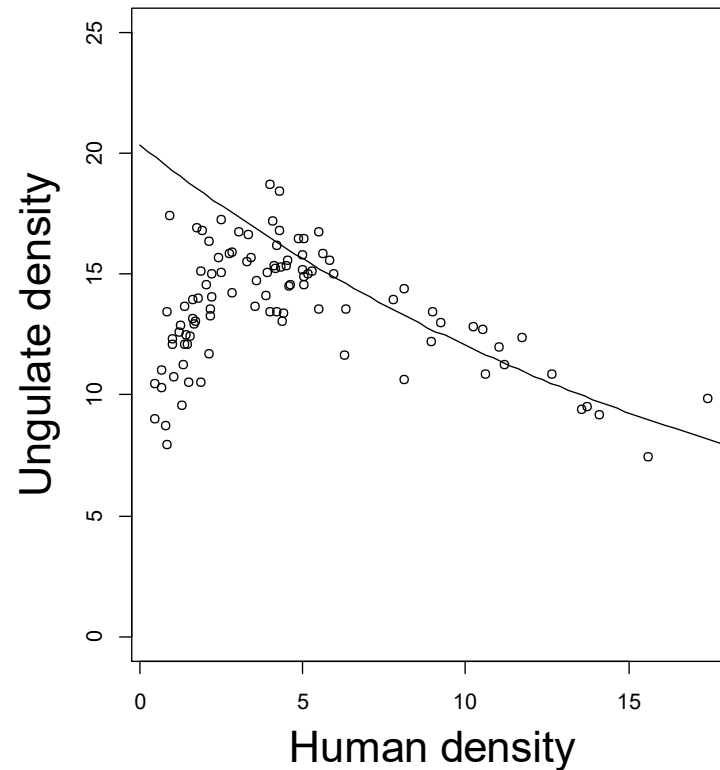
Today we have three main goals:

1. Fit a `glm()` and `piecewiseSEM`
2. Fit this in `lavaan` and `blavaan` (and export a JAGS file)
3. Fit this in JAGS and make predictions/figures



Fit a glm()

`glm(log(u) ~ w + h)`



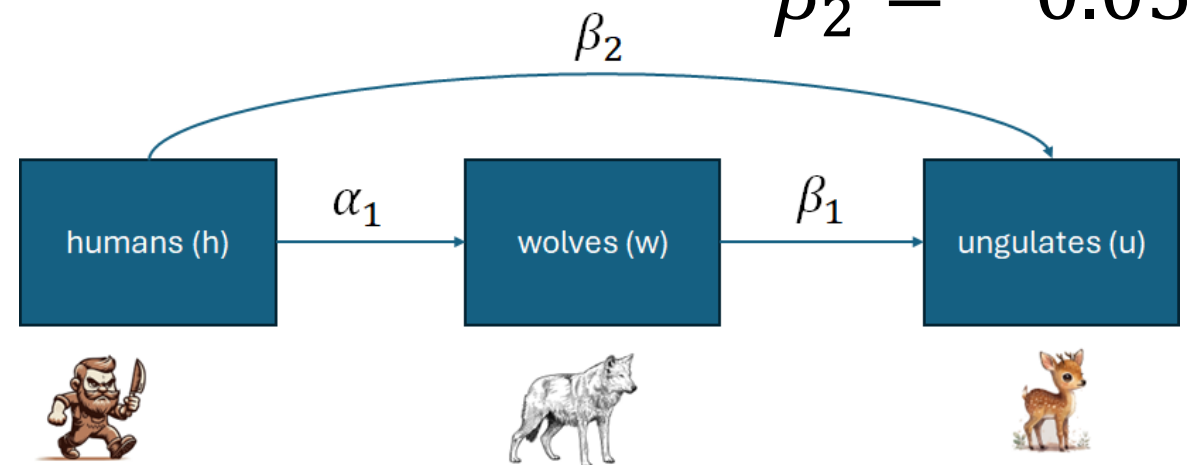
```
Call:
glm(formula = log(u) ~ w + h)
```

Coefficients:

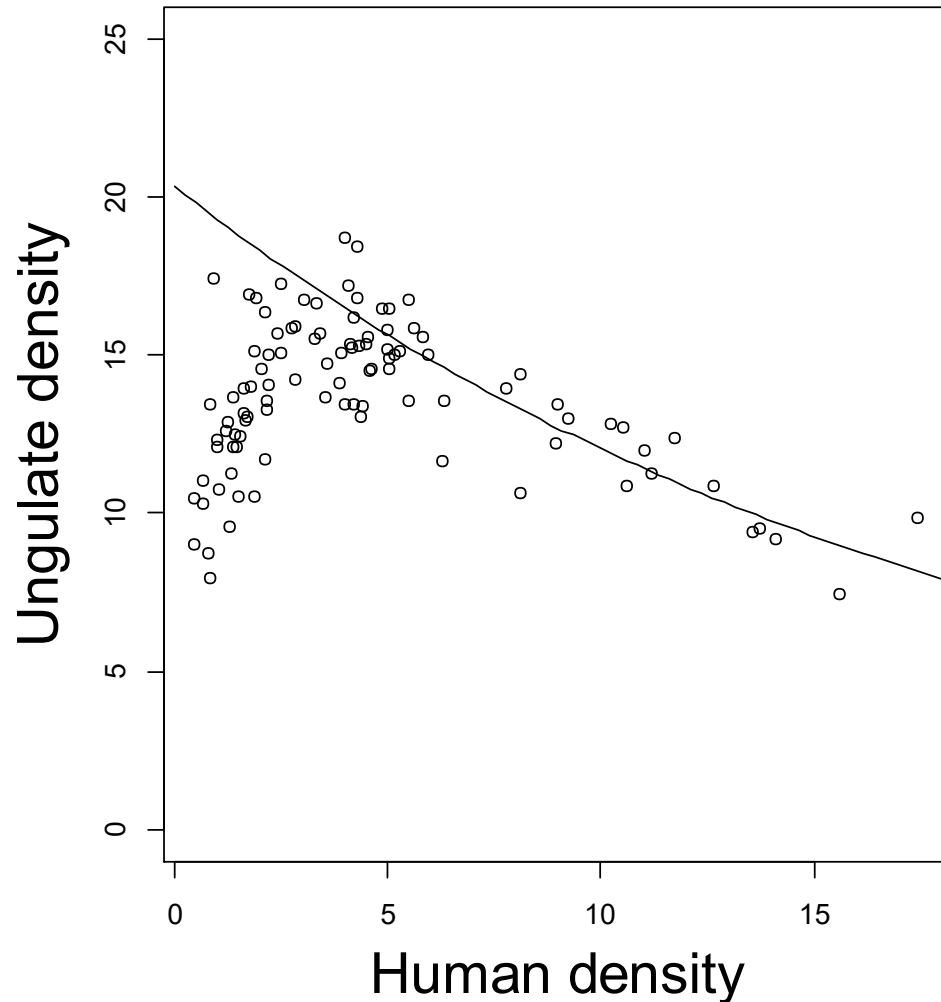
	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	3.012339	0.025943	116.11	<2e-16	***
w	-0.155041	0.009319	-16.64	<2e-16	***
h	-0.052451	0.003428	-15.30	<2e-16	***

$$\beta_1 = -0.15$$

$$\beta_2 = -0.05$$



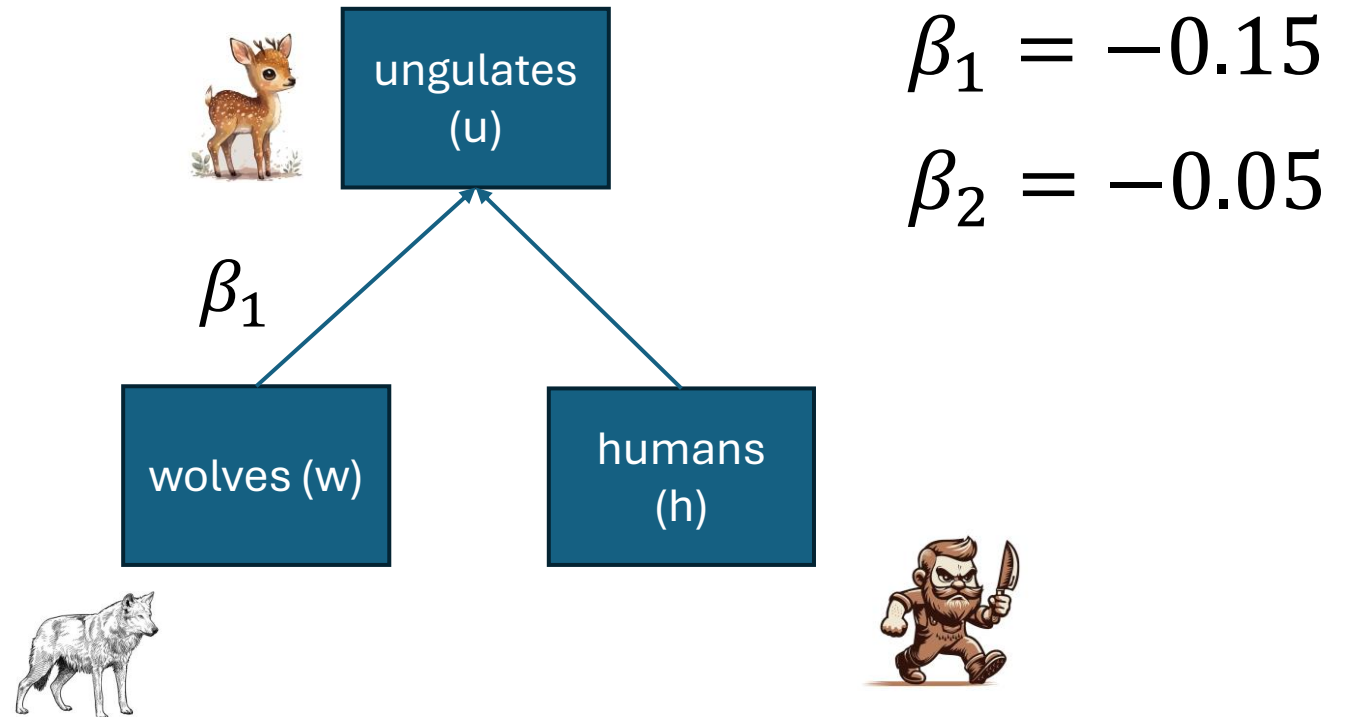
This is the relationship we'd predict if we held $w = 0$ and varied h



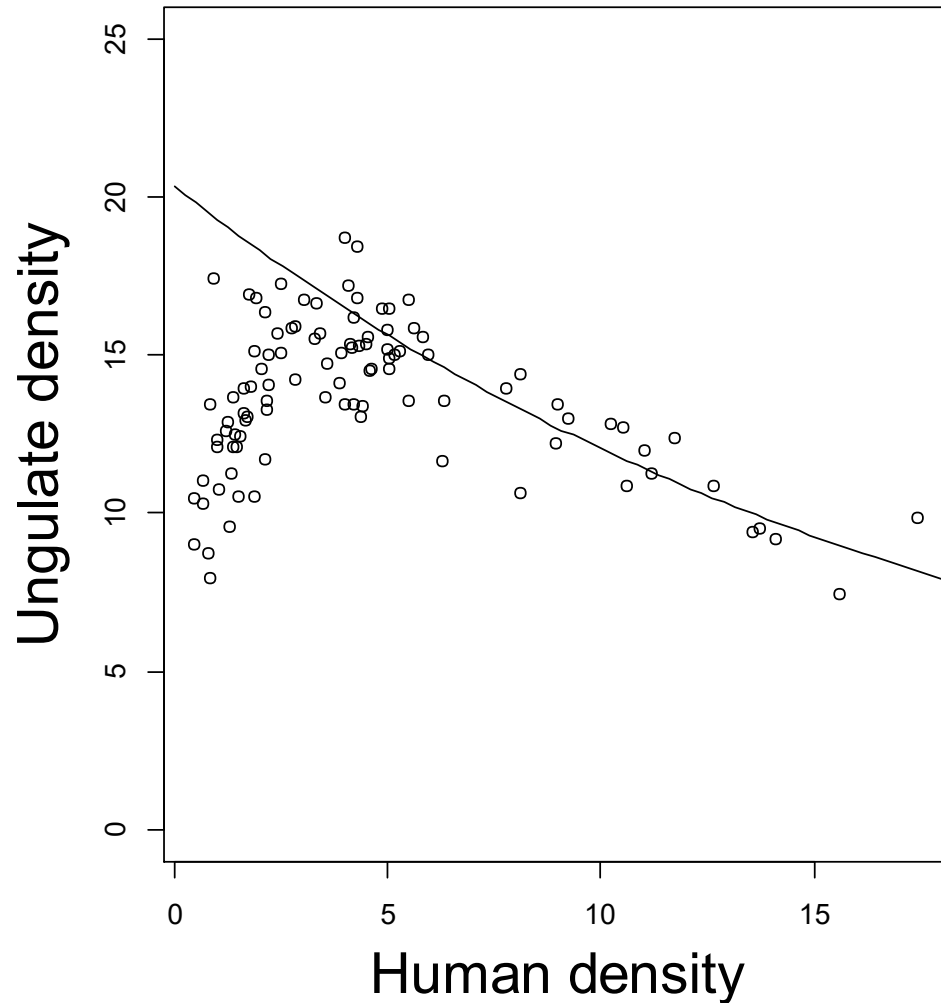
```
Call:
glm(formula = log(u) ~ w + h)

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  3.012339   0.025943  116.11  <2e-16 ***
w          -0.155041   0.009319  -16.64  <2e-16 ***
h          -0.052451   0.003428  -15.30  <2e-16 ***
---

```



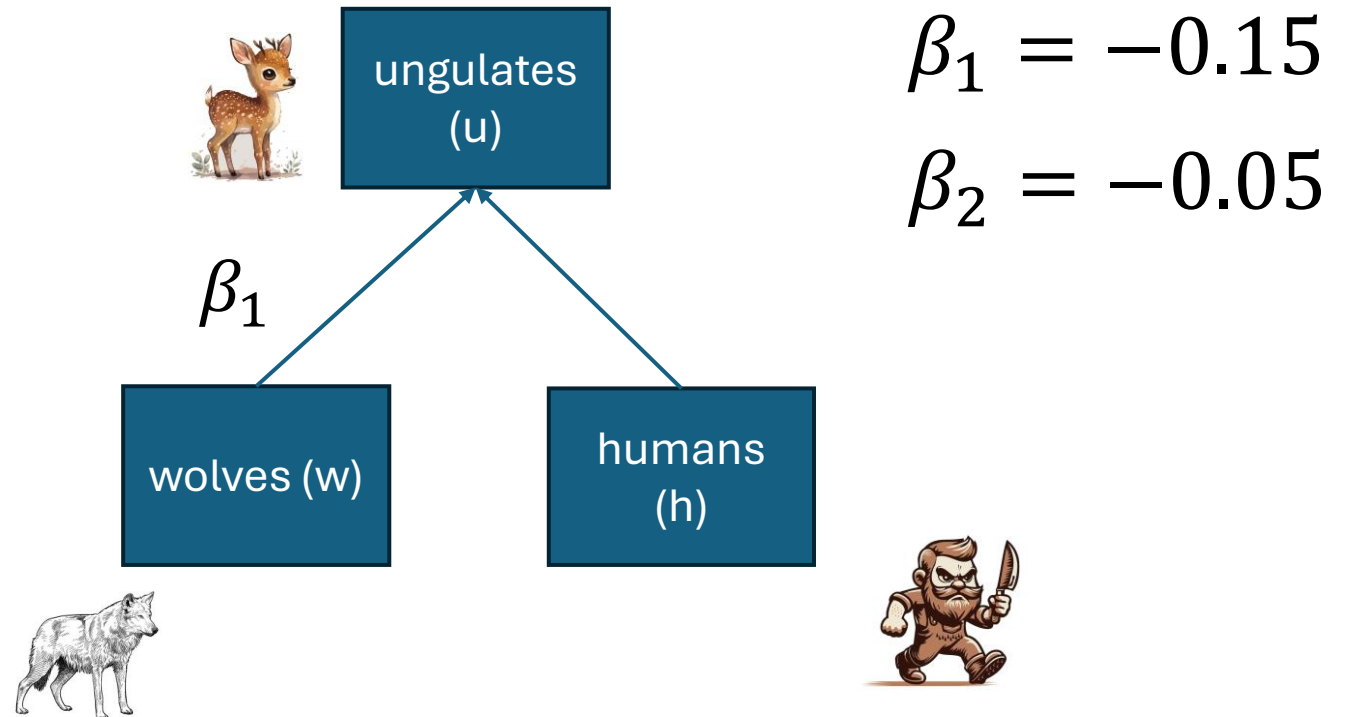
We can't hold wolves constant and vary h



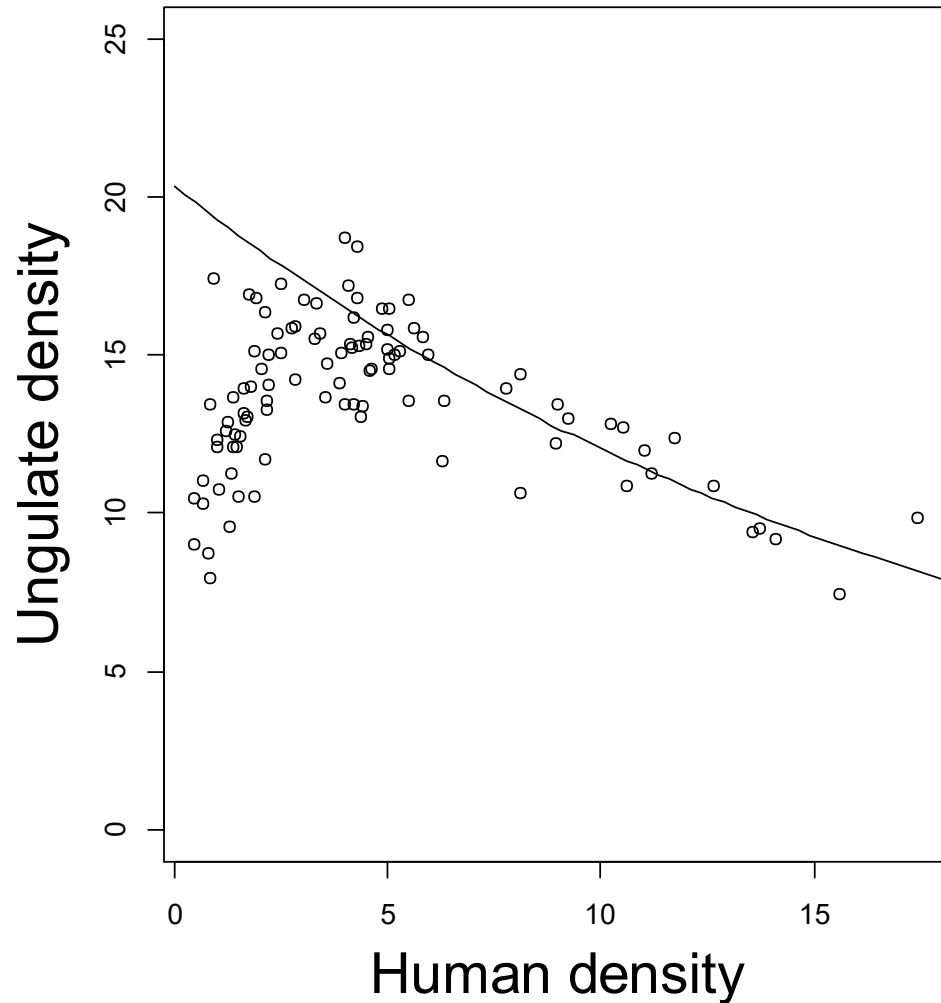
```
Call:
glm(formula = log(u) ~ w + h)

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  3.012339   0.025943  116.11  <2e-16 ***
w          -0.155041   0.009319  -16.64  <2e-16 ***
h          -0.052451   0.003428  -15.30  <2e-16 ***
---

```



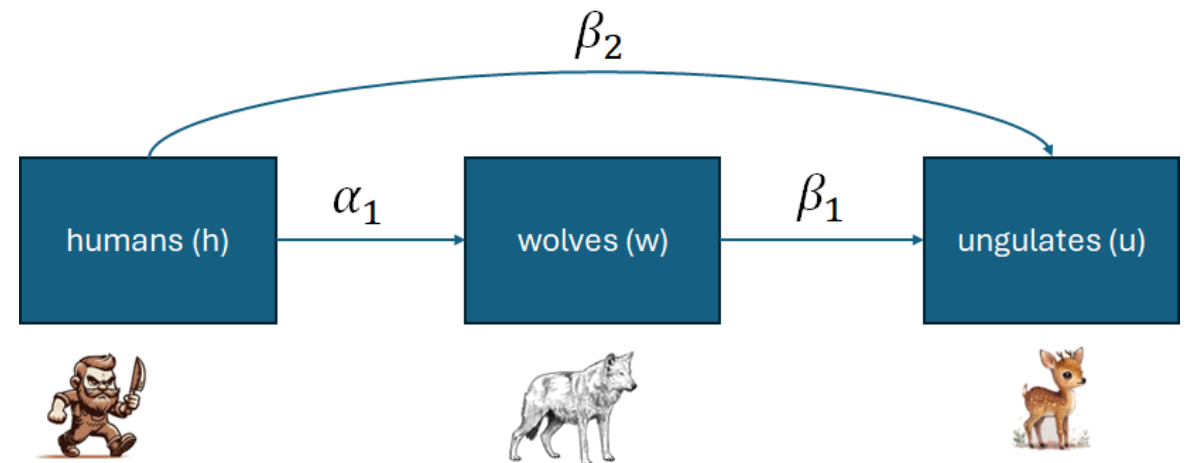
We can't hold w constant and vary h , because h affects w !



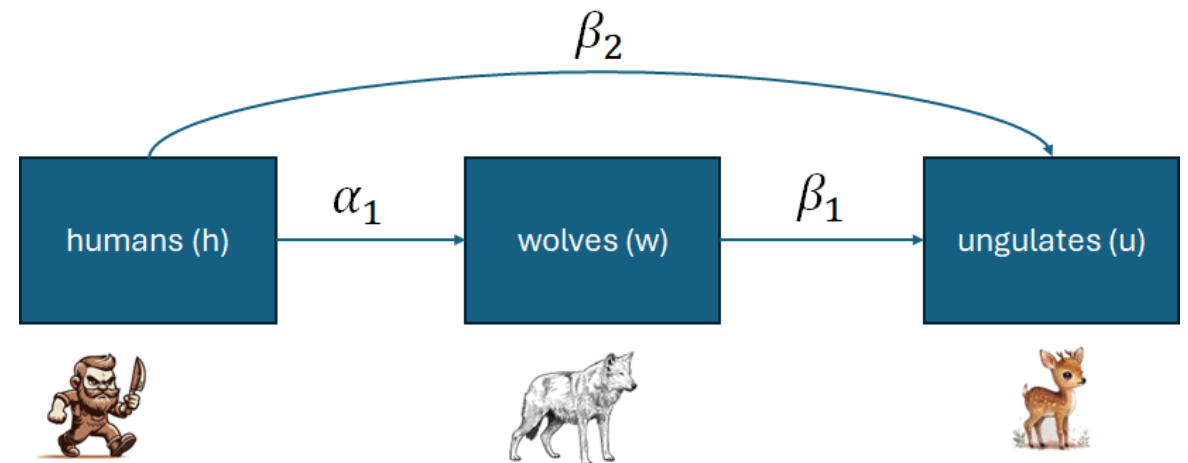
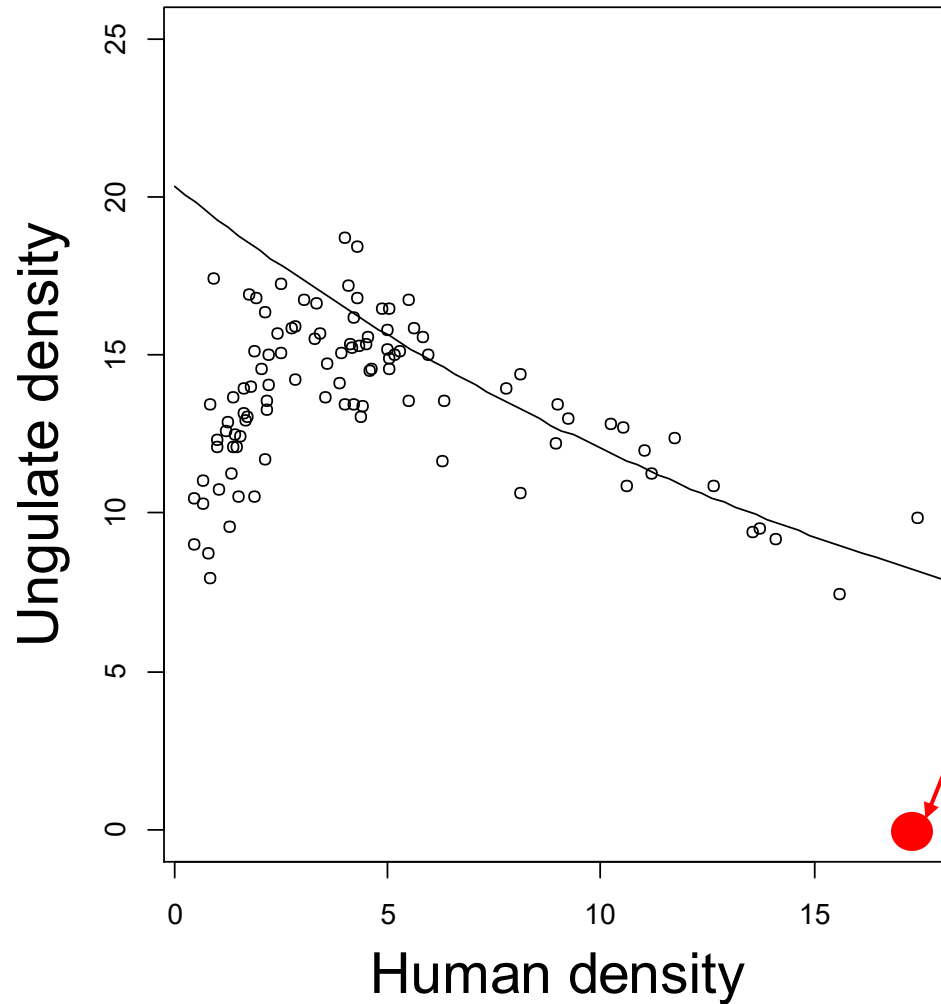
```
Call:
glm(formula = log(u) ~ w + h)

Coefficients:
(Intercept)  Estimate Std. Error t value Pr(>|t|)
w            -0.155041  0.009319  -16.64  <2e-16 ***
h            -0.052451  0.003428  -15.30  <2e-16 ***
---

```

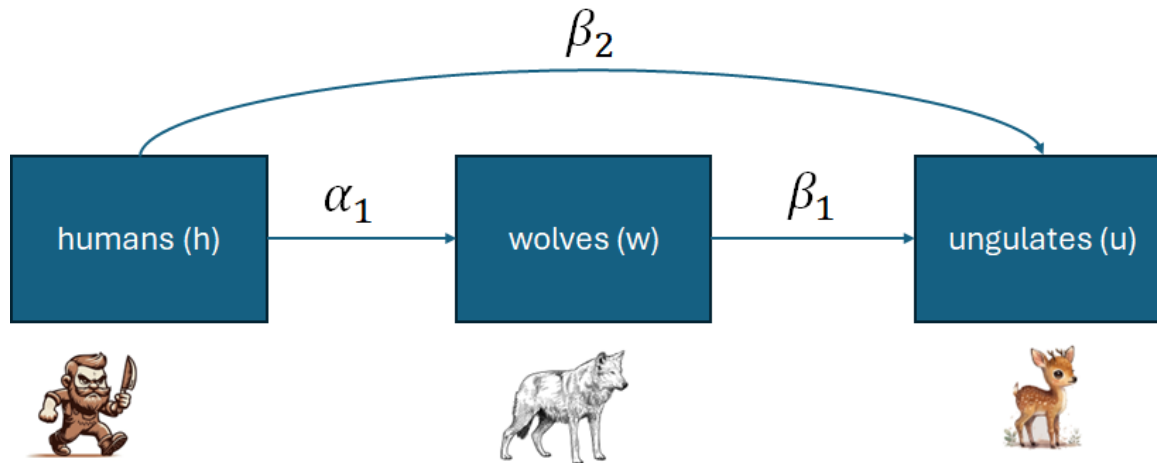


This also helps avoid silly predictions (what if h and w were high?)



Fit in piecewiseSEM (we can add link functions)

```
# ~~~~~  
# Third, let's explore piecewiseSEM  
# ~~~~~  
m.psem <- psem(  
  glm(log(w) ~ h, data = data),  
  glm(log(u) ~ h + w, data = data),  
  data = data  
)  
summary(m.psem)
```



Structural Equation Model of m.psem

Call:

```
log(w) ~ h  
log(u) ~ h + w
```

AIC

-171.169

Tests of directed separation:

	Independ.Claim	Test.Type	DF	Crit.Value	P.Value
log(w) ~ w + ...	coef 97			1.3539	0.1789
log(u) ~ log(w) + ...	coef 96			-1.1556	0.2507

--

Global goodness-of-fit:

Chi-Squared = 3.253 with P-value = 0.197 and on 2 degrees of freedom
Fisher's C = 6.208 with P-value = 0.184 and on 4 degrees of freedom

Coefficients:

Response	Predictor	Estimate	Std.Error	DF	Crit.Value	P.Value	Std.Estimate	
log(w)	h	-0.7445	0.0069	98	-108.5764	0	-0.9959	***
log(u)	h	-0.0525	0.0034	97	-15.2989	0	-1.0248	***
log(u)	w	-0.1550	0.0093	97	-16.6368	0	-1.1144	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05

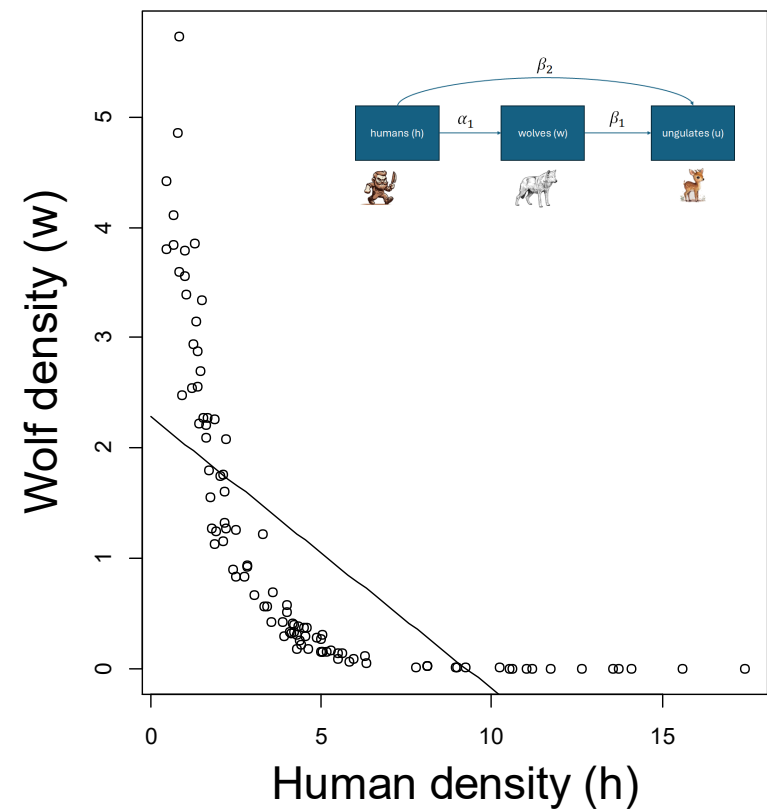
Individual R-squared:

Response	method	R.squared
w	none	0.99
u	none	0.76

And we accurately recover our parameters!

Fit in lavaan

```
m.lavaan <- sem(model = '
    w ~ 1 + h
    u ~ 1 + w + h', data = data)
summary(m.lavaan)
```



```
> summary(m.lavaan)
lavaan 0.6-18 ended normally after 1 iteration

Estimator              ML
Optimization method     NLMINB
Number of model parameters 7

Number of observations    100

Model Test User Model:

Test statistic           0.000
Degrees of freedom       0

Parameter Estimates:

Standard errors              Standard
Information                  Expected
Information saturated (h1) model Structured

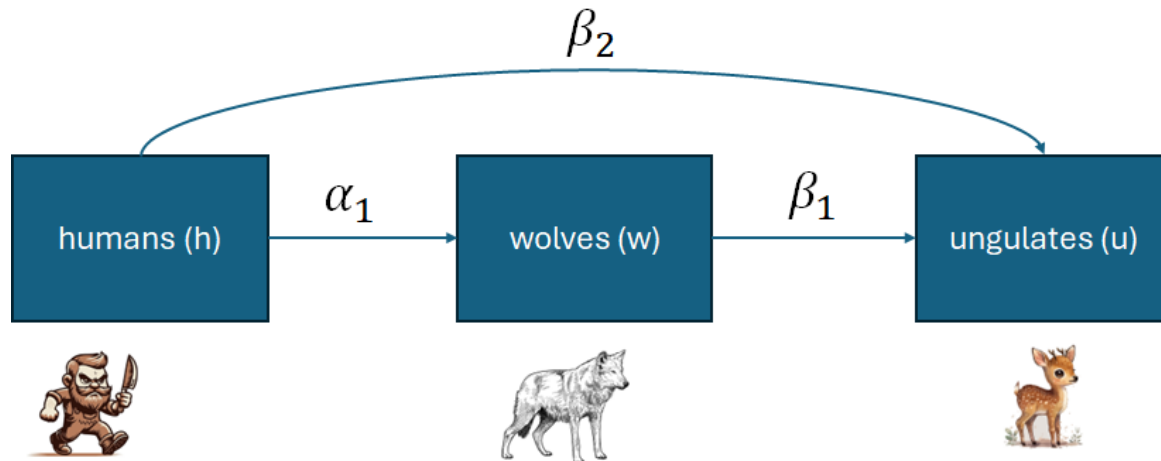
Regressions:
      Estimate Std.Err z-value P(>|z|)
w ~
  h          -0.247   0.027  -9.039   0.000
u ~
  w          -1.923   0.126 -15.258   0.000
  h          -0.646   0.046 -13.941   0.000

Intercepts:
      Estimate Std.Err z-value P(>|z|)
.w           2.284   0.159  14.353   0.000
.u           18.783   0.351  53.533   0.000

Variances:
      Estimate Std.Err z-value P(>|z|)
.w           1.016   0.144   7.071   0.000
.u           1.614   0.228   7.071   0.000
```

Fit in blavaan

```
library(runjags)
m.blavaan <- bsem(model = '
  w ~ 1 + h
  u ~ 1 + w + h', data = data,
  target = 'jags',
  mcmcfile = T)
summary(m.blavaan)
```



```
> summary(m.blavaan)
blavaan 0.5.6 ended normally after 10000 iterations
```

Estimator	BAYES	
Optimization method	MCMC	
Number of model parameters	7	
Number of observations	100	
Number of missing patterns	1	
Statistic Value	MargLogLik -339.545	PPP 0.002

Parameter Estimates:

Regressions:

	Estimate	Post.SD	pi.lower	pi.upper	Rhat	Prior
w ~						
h	-0.247	0.028	-0.301	-0.192	1.000	dnorm(0,1e-2)
u ~						
w	-1.924	0.129	-2.183	-1.676	1.002	dnorm(0,1e-2)
h	-0.646	0.047	-0.736	-0.553	1.002	dnorm(0,1e-2)

Intercepts:

	Estimate	Post.SD	pi.lower	pi.upper	Rhat	Prior
.w	2.285	0.160	1.977	2.604	1.000	dnorm(0,1e-3)
.u	18.784	0.358	18.068	19.465	1.002	dnorm(0,1e-3)

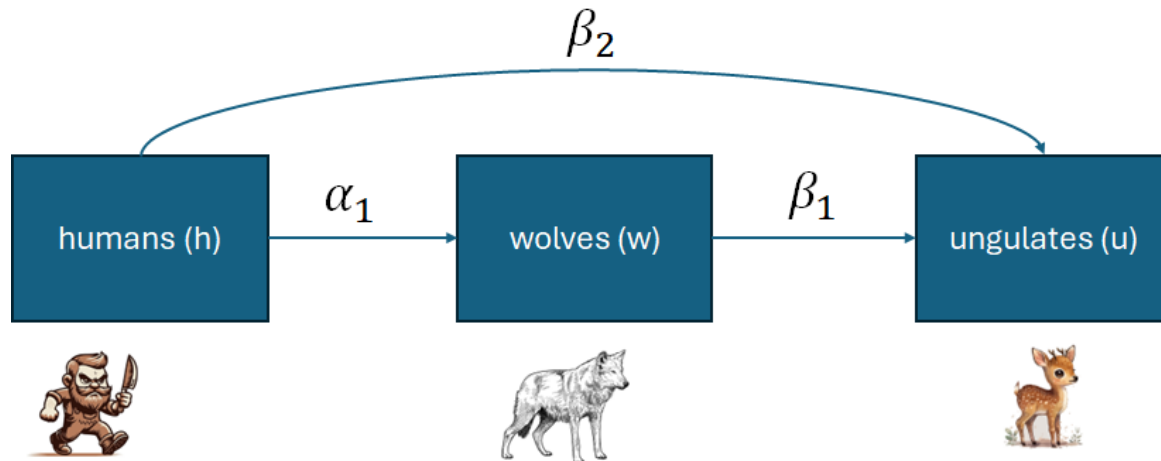
Variances:

	Estimate	Post.SD	pi.lower	pi.upper	Rhat	Prior
.w	1.046	0.151	0.772	1.355	1.000	dgamma(1,.5)[prec]
.u	1.677	0.244	1.230	2.163	1.000	dgamma(1,.5)[prec]

Note the similarities between blavaan and lavaan

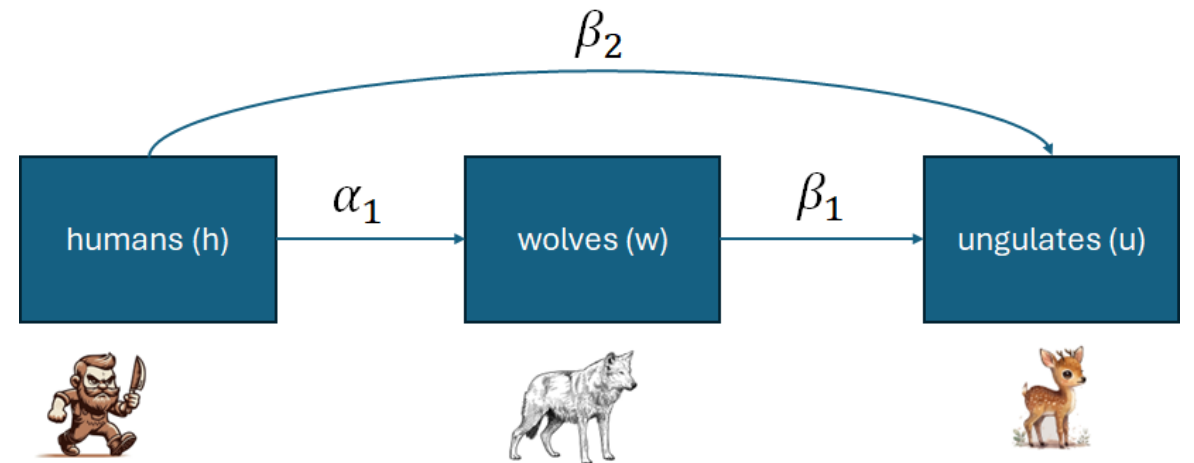
```
library(runjags)
m.blavaan <- bsem(model = '
    w ~ 1 + h
    u ~ 1 + w + h', data = data,
    target = 'jags',
    mcmcfile = T)
summary(m.blavaan)
```

```
m.lavaan <- sem(model = '
    w ~ 1 + h
    u ~ 1 + w + h', data = data)
summary(m.lavaan)
```



mcmcfile = T saves a working model in Stan or JAGS

```
library(runjags)
m.blavaan <- bsem(model = '
  w ~ 1 + h
  u ~ 1 + w + h', data = data,
  target = 'jags',
  mcmcfile = T)
summary(m.blavaan)
```



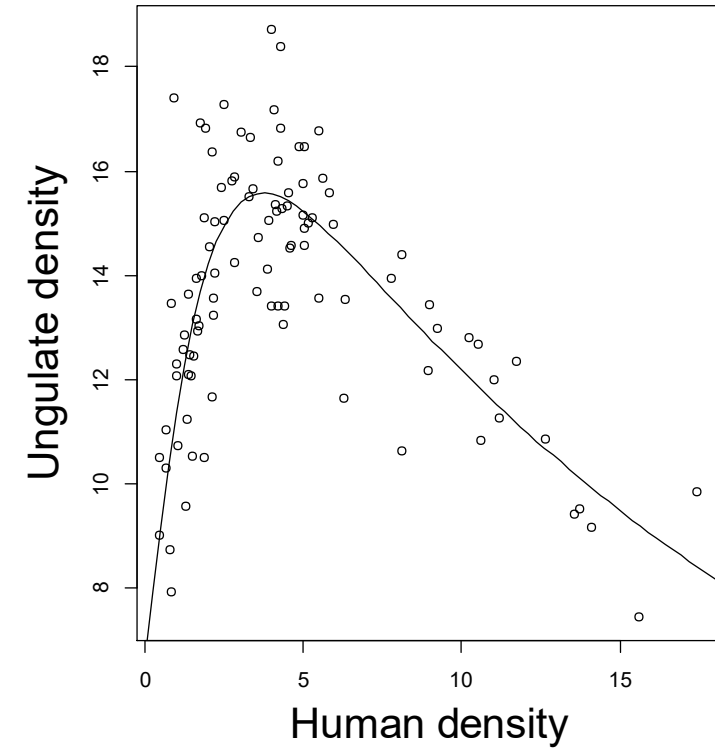
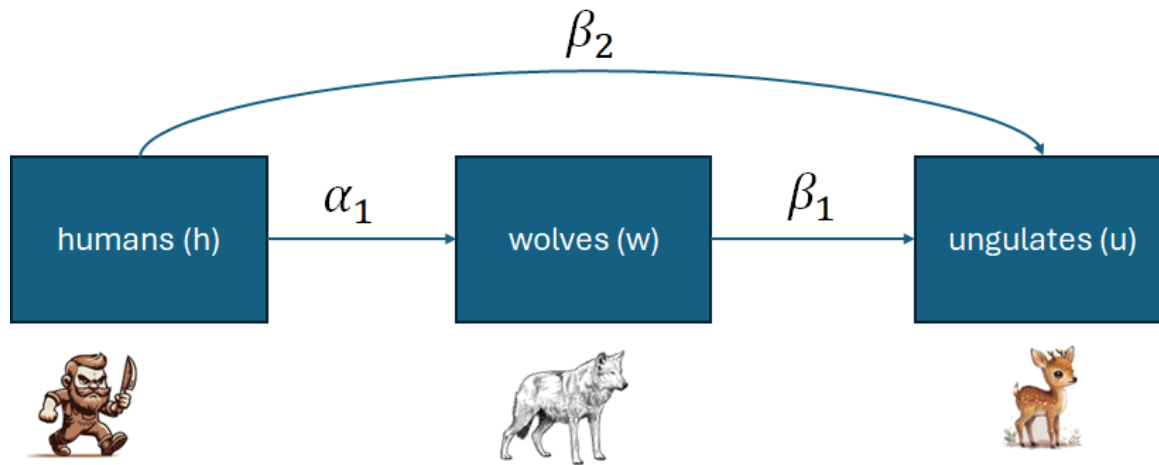
lavExport (folder in your working directory)

**Let's take a quick break, then we'll go to our R script,
work through our own custom JAGS model,
and explore our first counter-factual*.**

***i.e., what would've happened if...?**

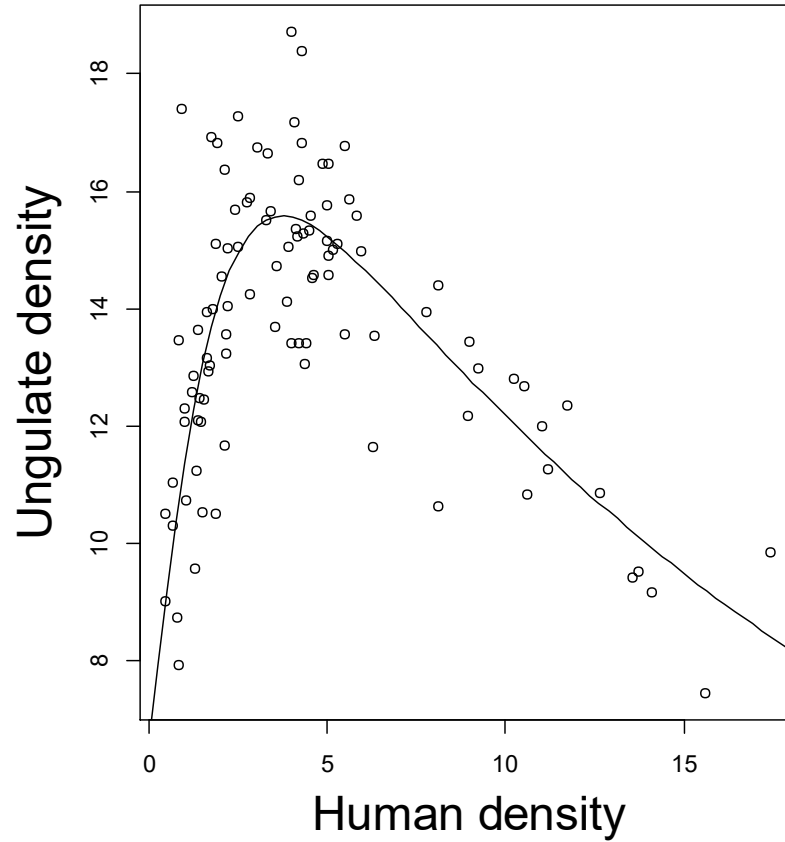
lavExport (folder in Documents or your working directory)

Take-home 1: complex paths produce complex signals



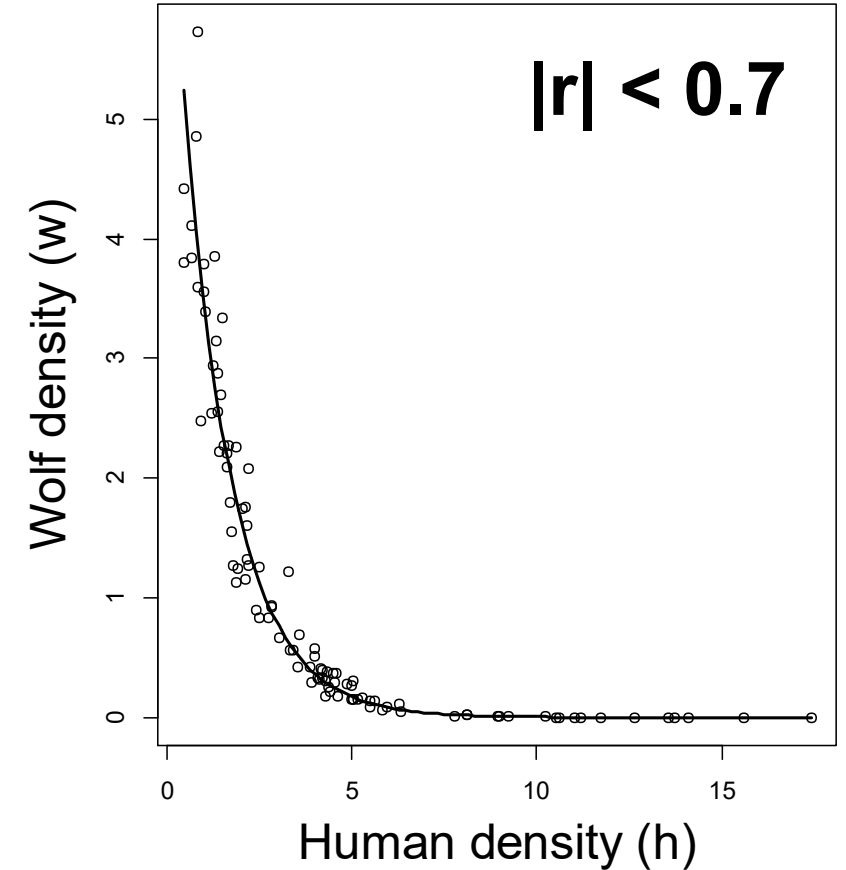
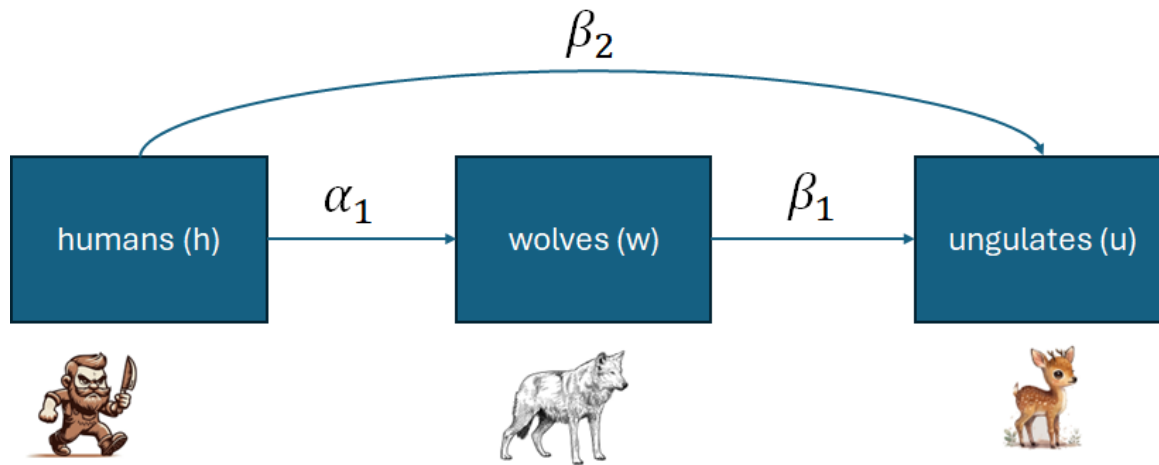
***Almost everything (worth studying) is the result of a complex path!**

Take-home 2: univariate relationships can be misleading!



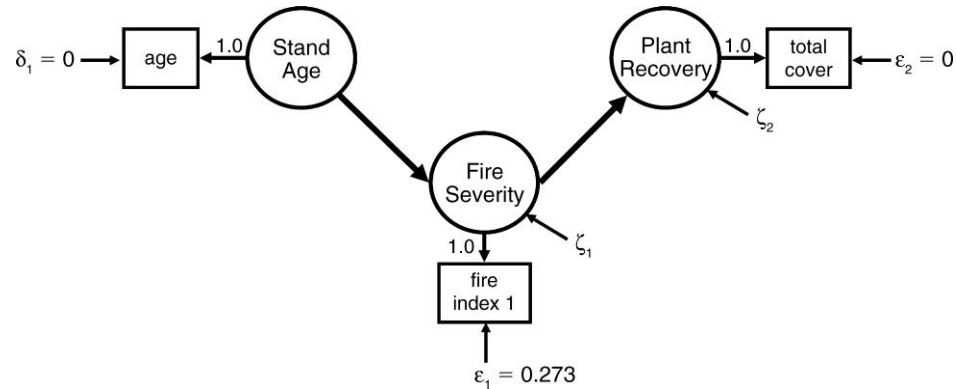
$\text{Im}(u \sim h)$

Take-home 3: Causal paths lead to multicollinearity



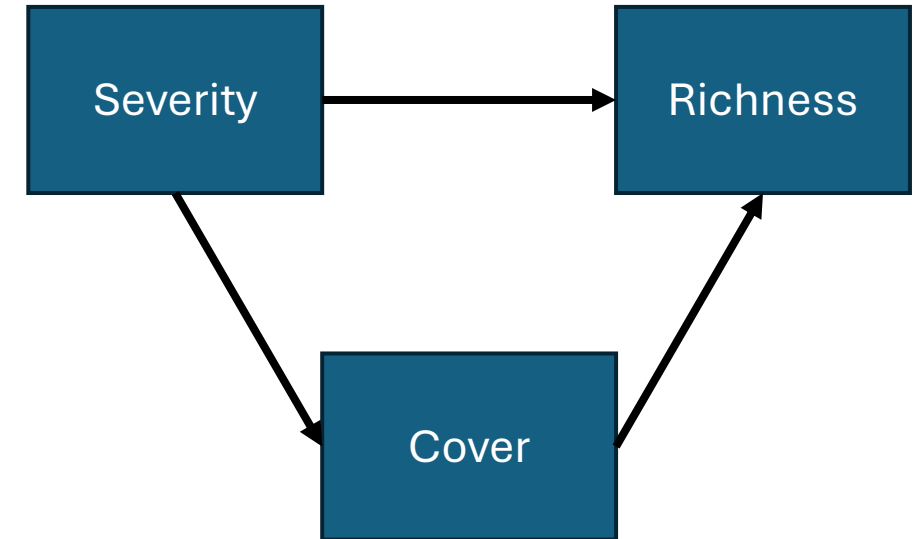
We can model those pathways to obtain more accurate predictive inference

Another example(s): the Keeley data [in 'pssem' package]



Grace et al. (2010) *Ecological Monographs*

Grace & Keeley (2006) *Ecological Applications*

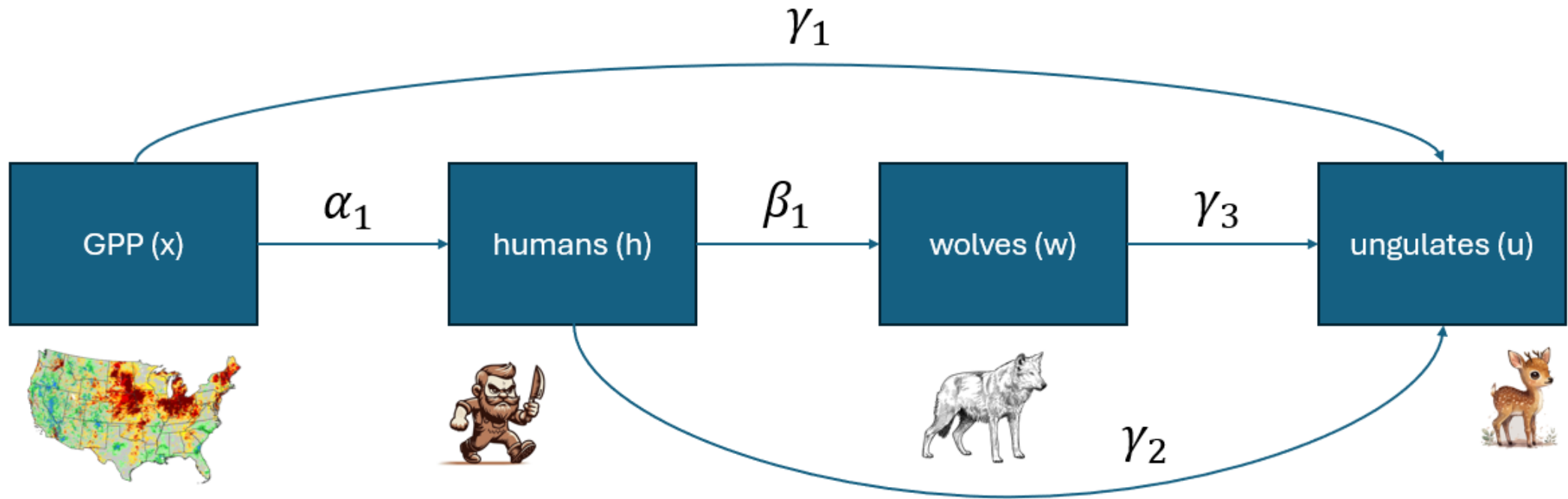


Some notes from Jon Lefcheck (jonlecheck.net)

https://www.youtube.com/watch?v=VT-gw_VVP1E

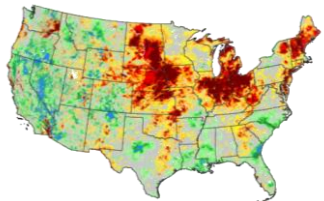
https://jslefcche.github.io/sem_book/

Following slides correspond to the supplementary script

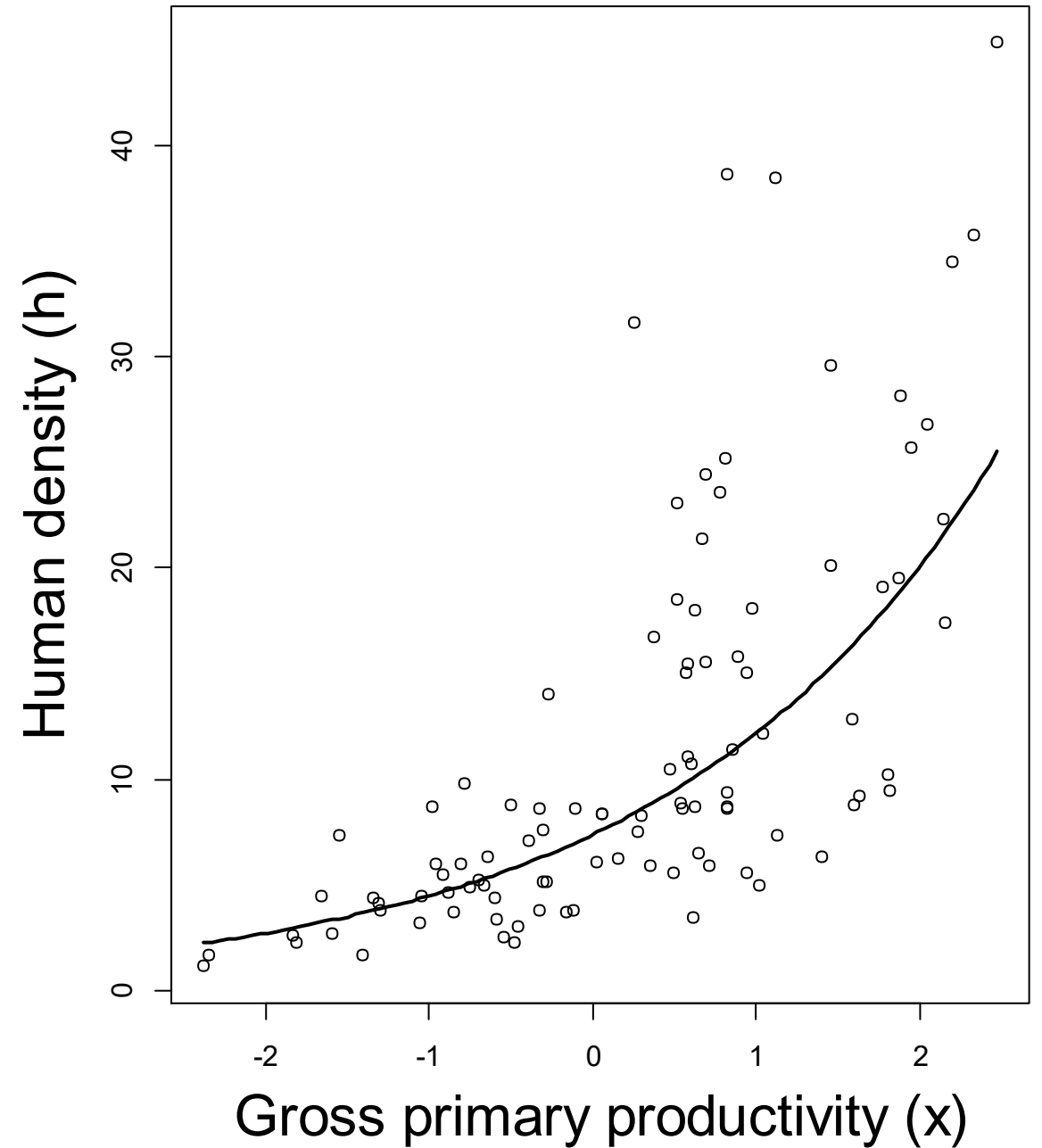
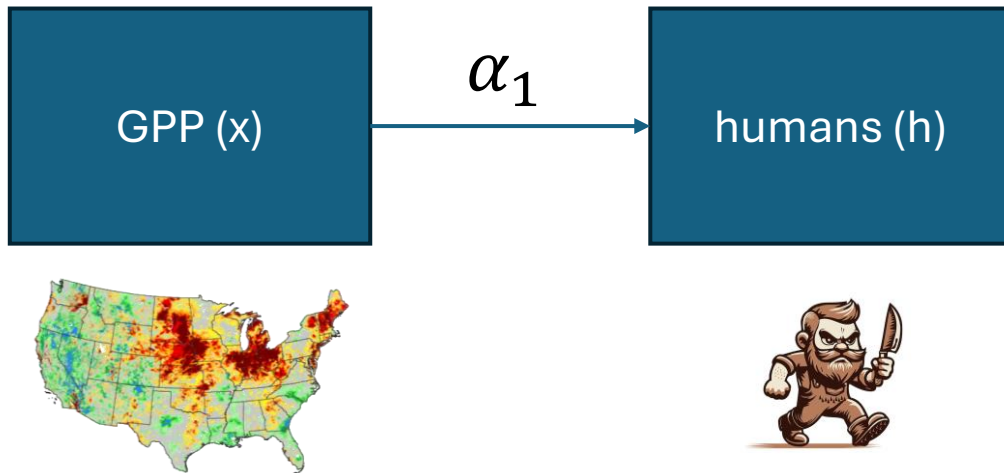


Simulating the data: GPP

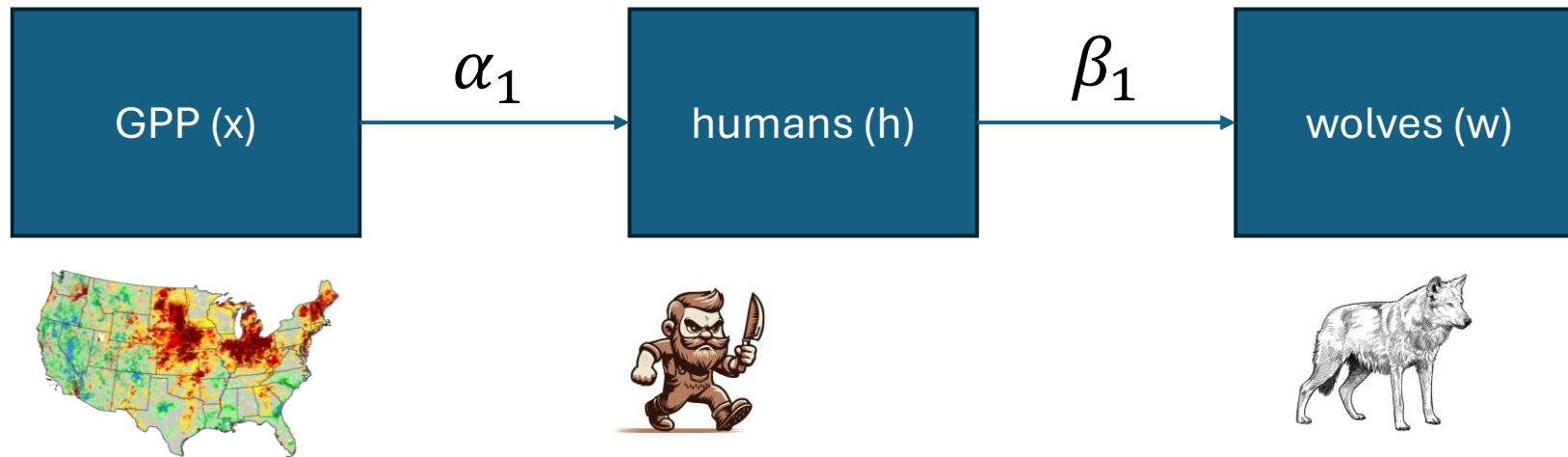
GPP (x)



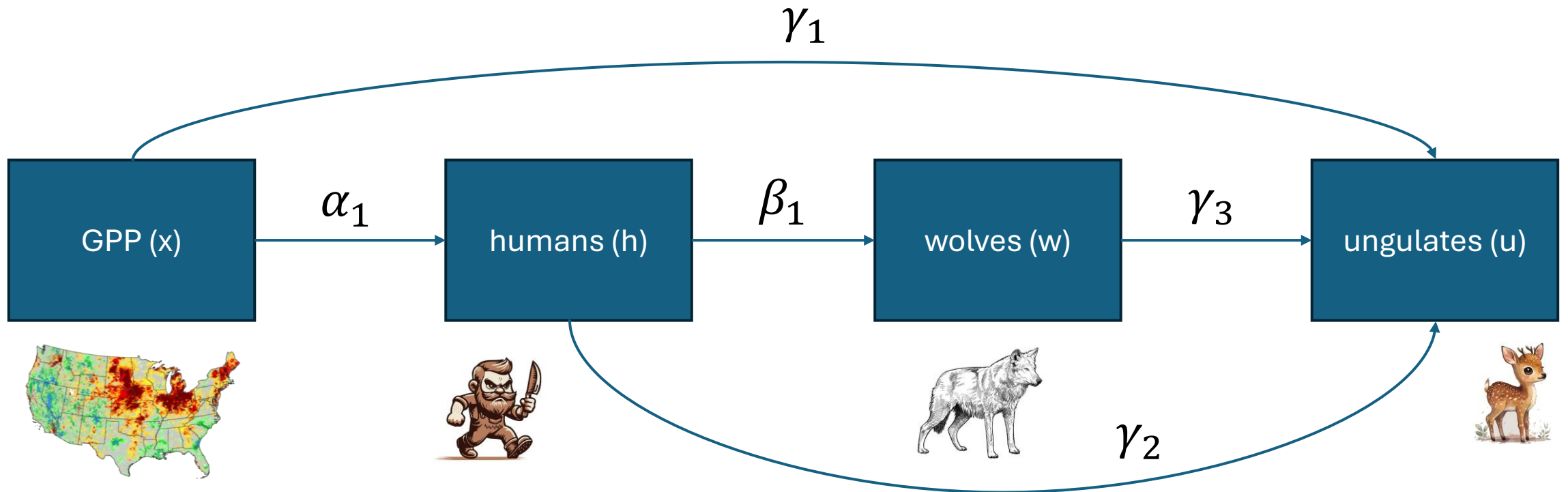
Simulating the data: humans



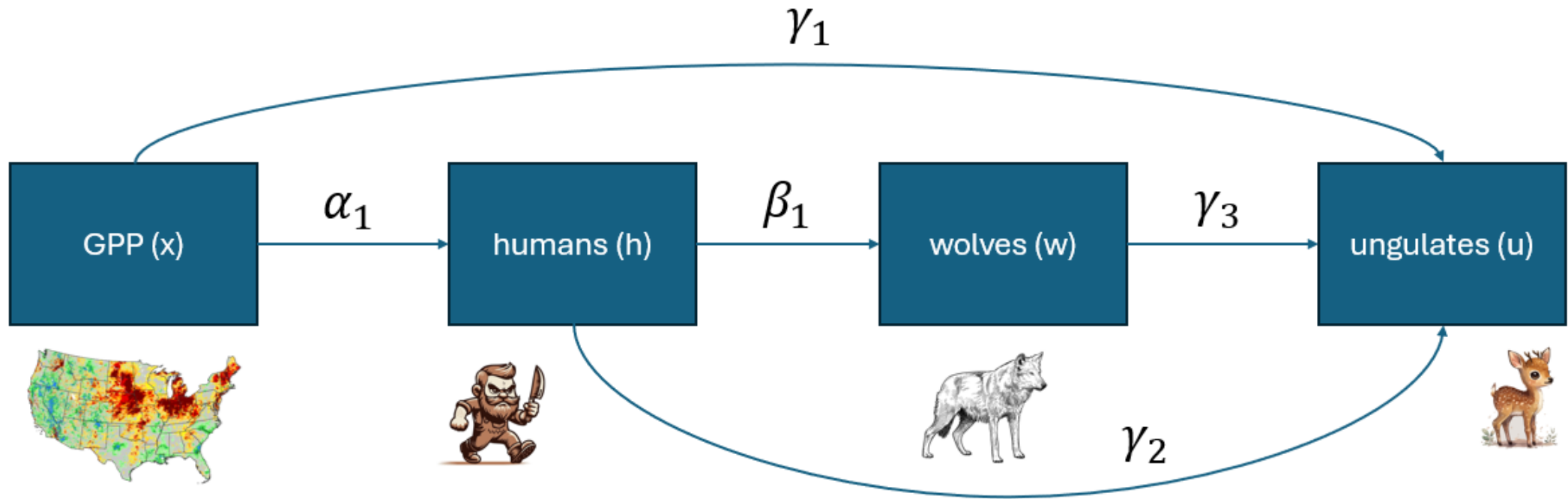
Simulating the data: wolves



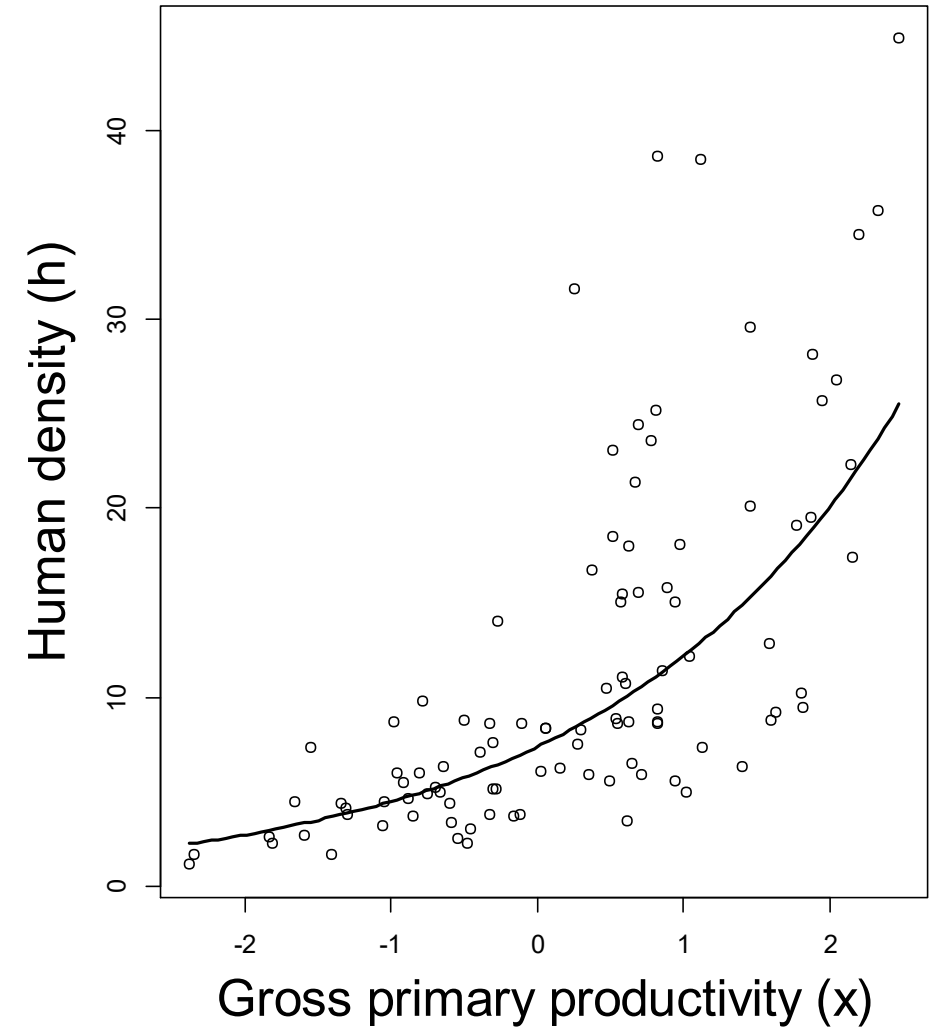
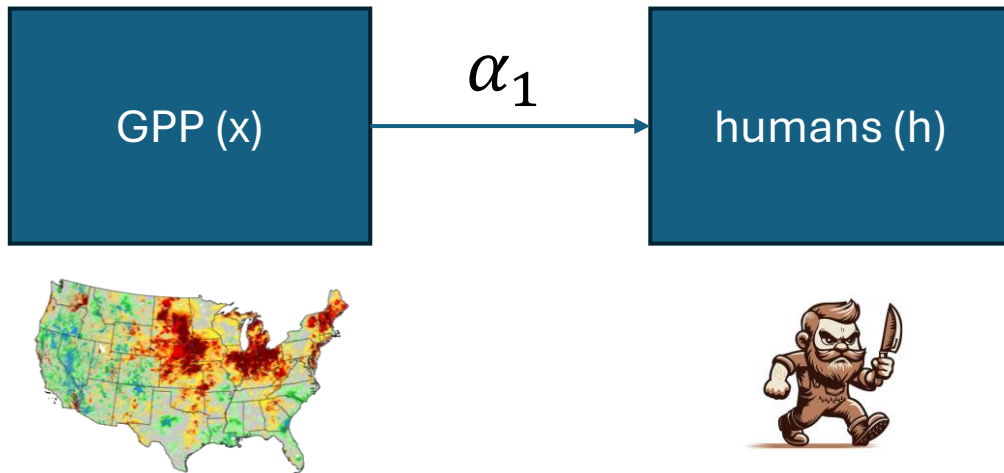
Simulating the data: ungulates



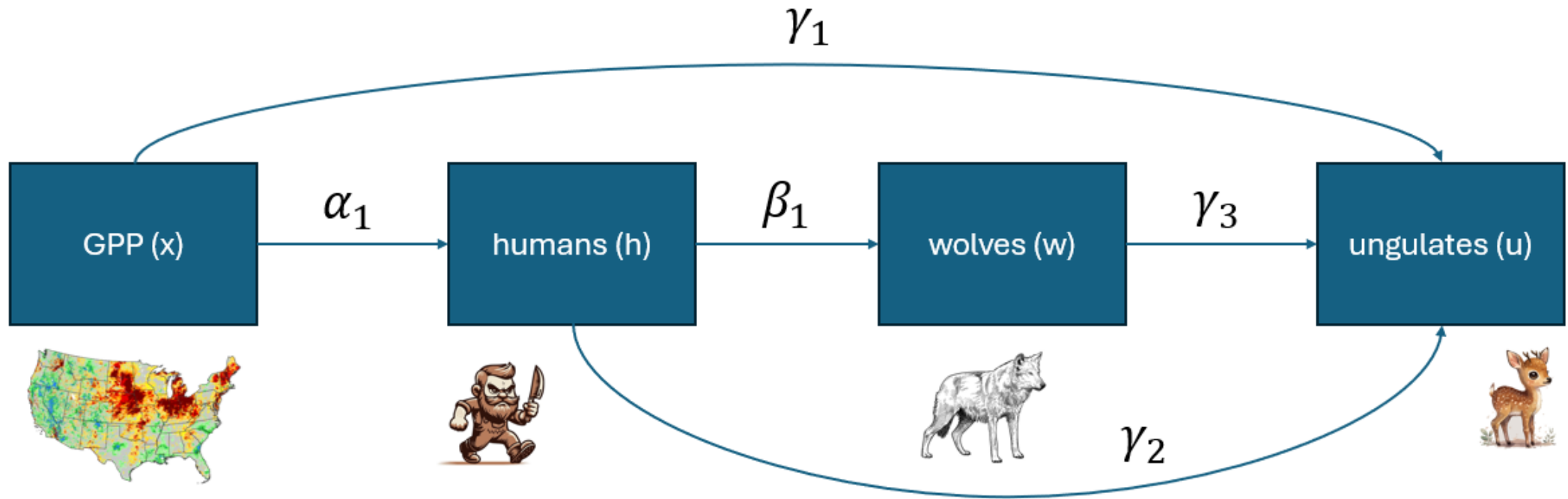
So what does this all mean?!



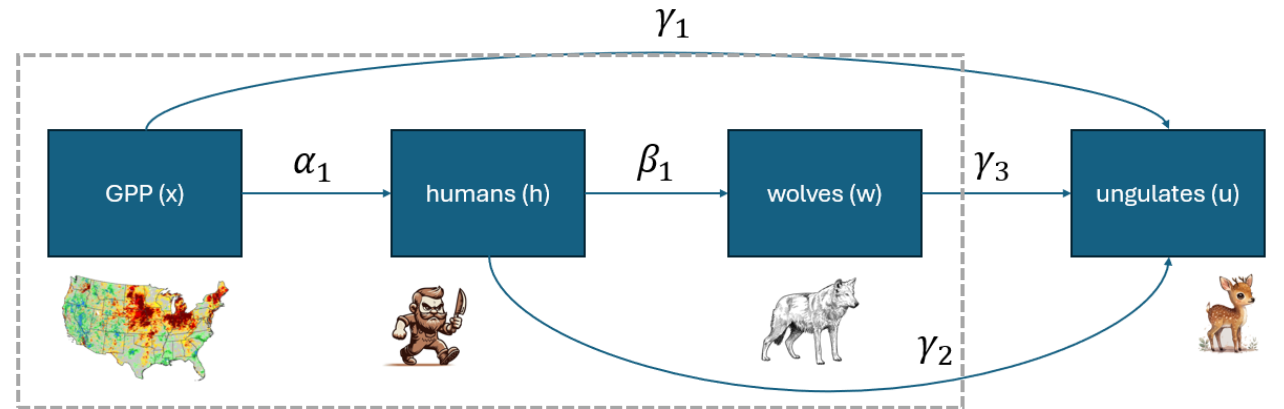
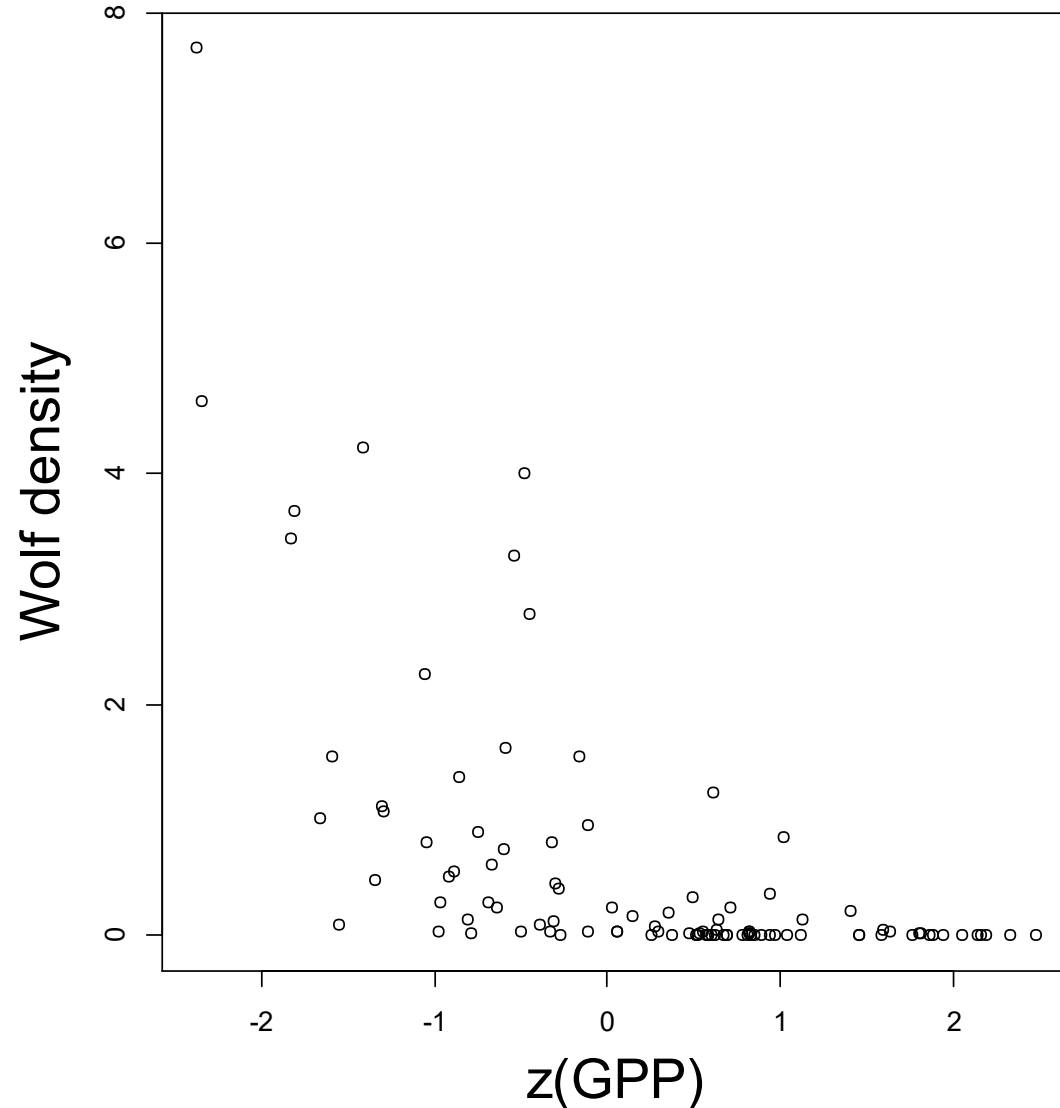
Simulating univariate relationships is simple



Do the presence of direct and indirect effects muddle things up?

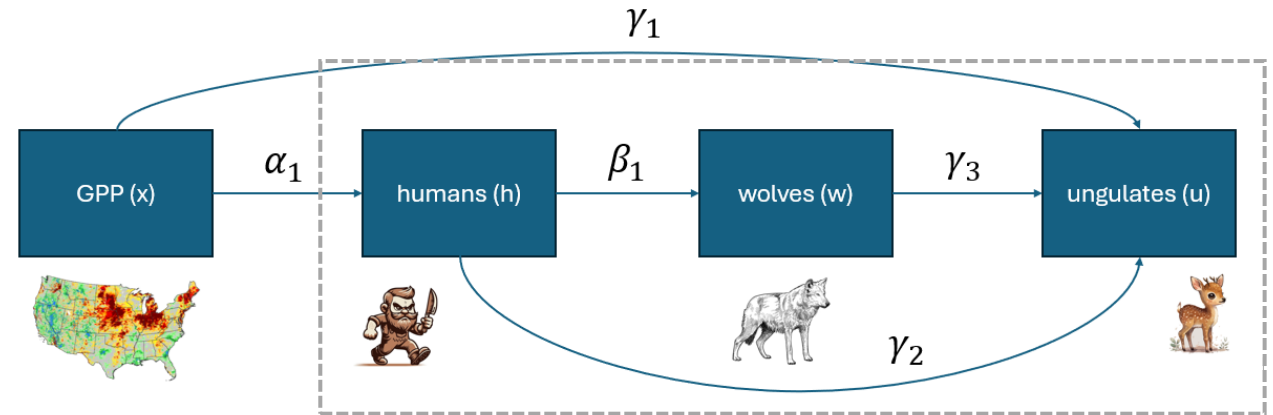
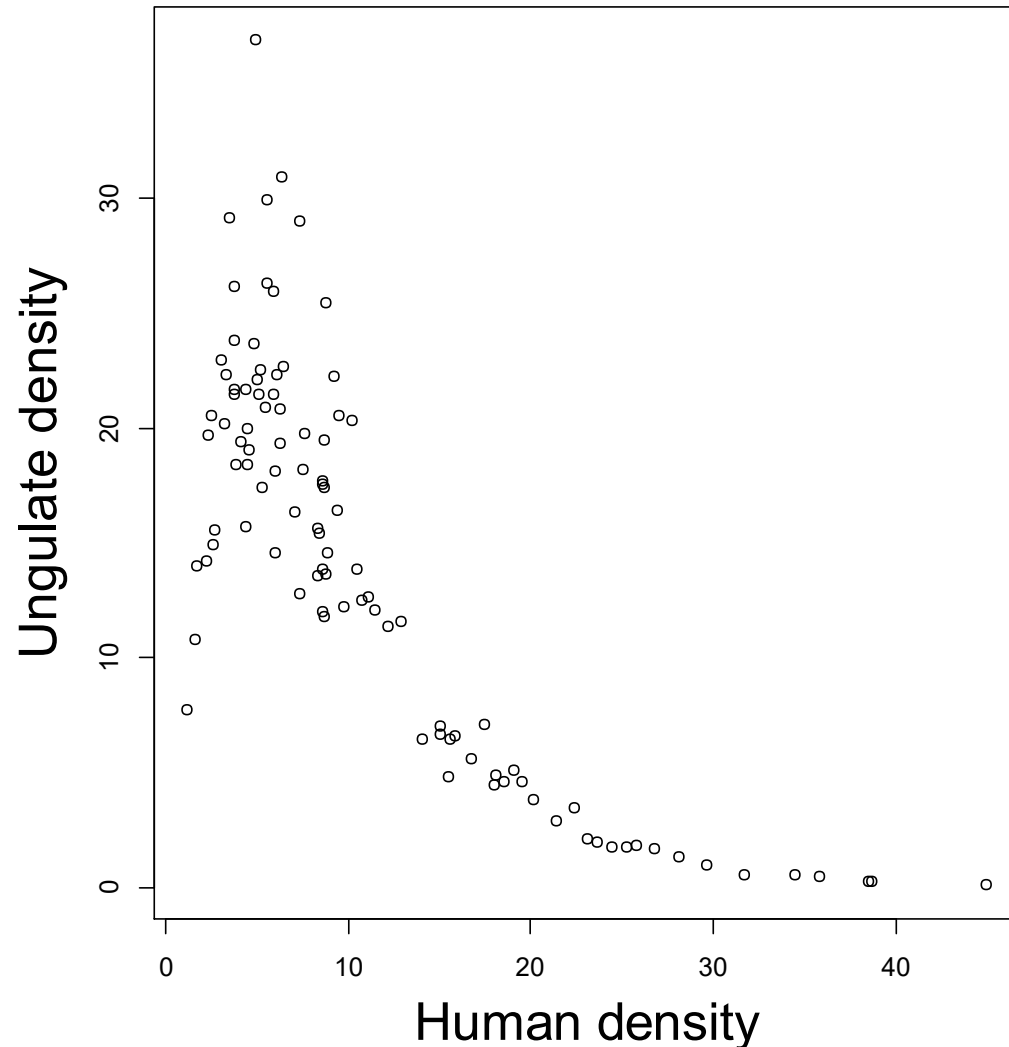


Here is the simulated relationship between wolves and GPP



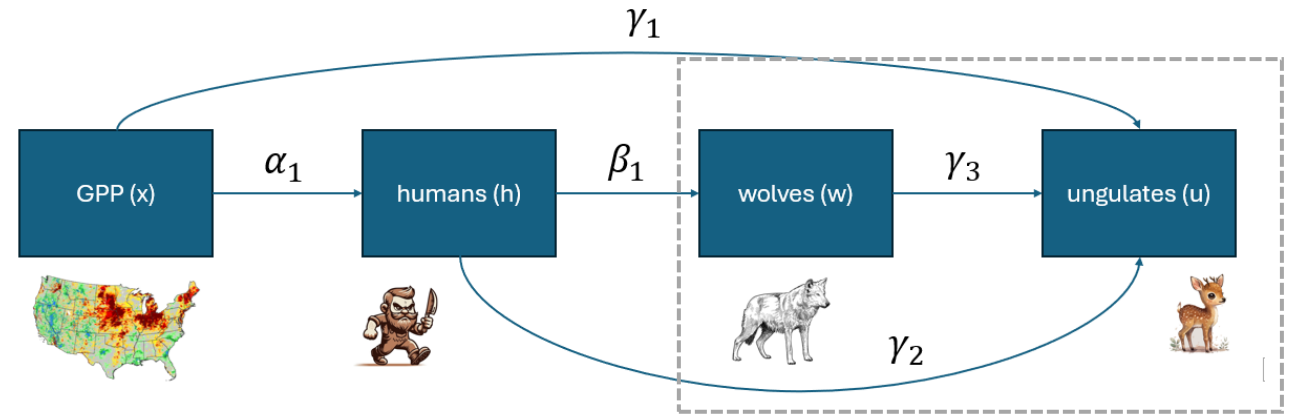
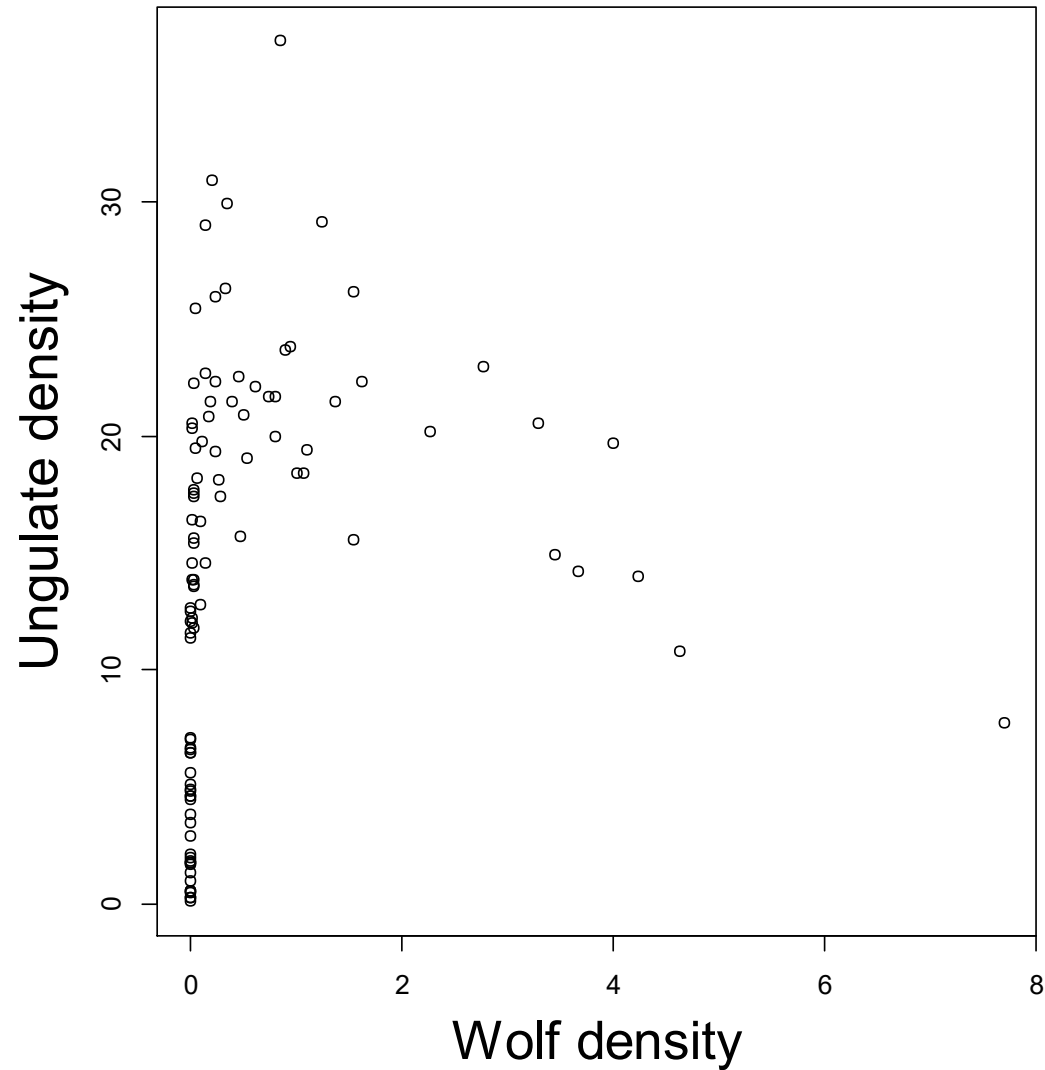
It's negative, not because GPP is bad for wolves, but b/c GPP means more humans. Humans are bad for wolves.

Here is the simulated relationship between ungulates and humans



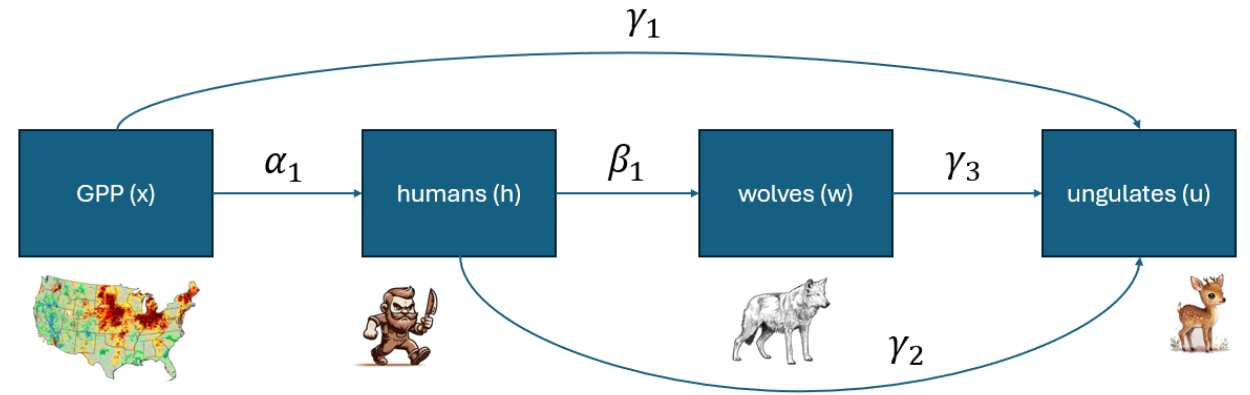
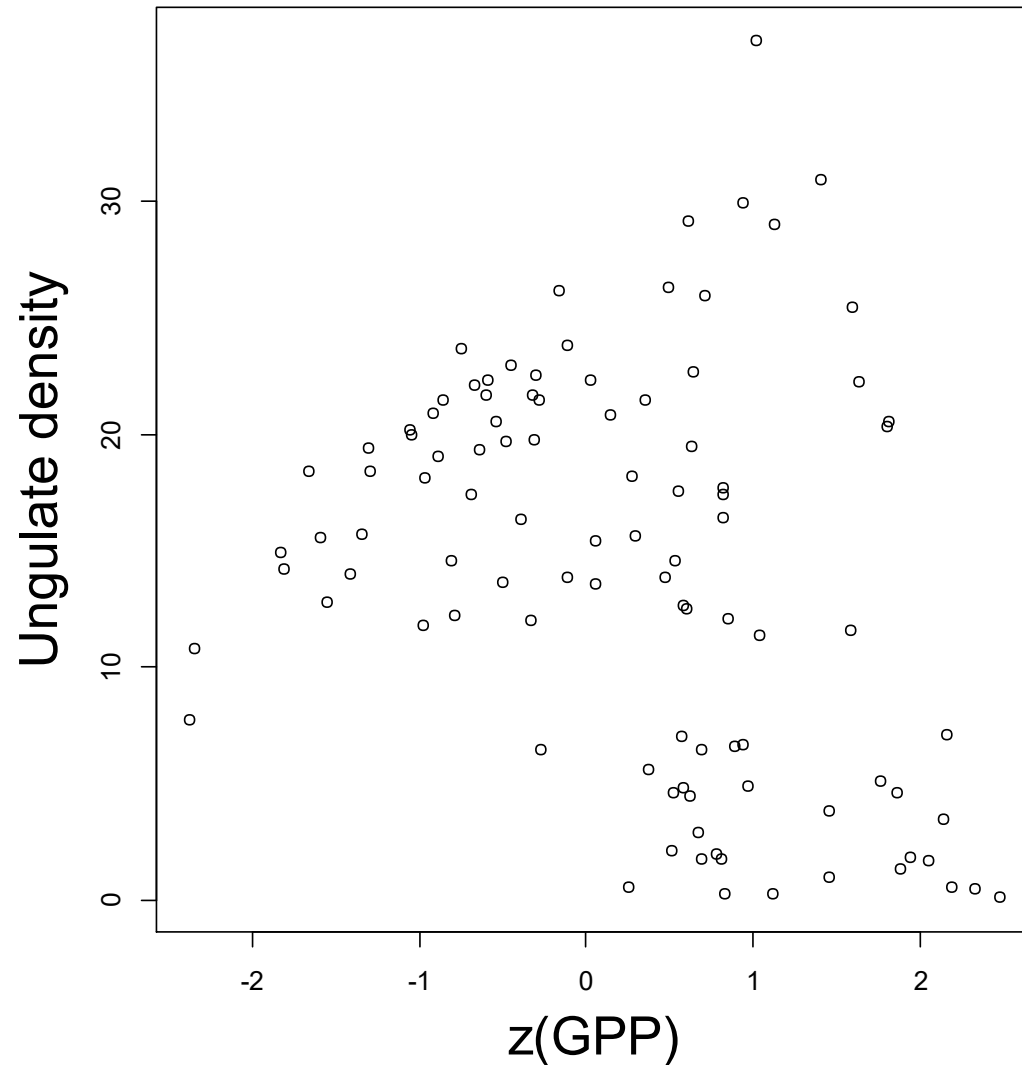
It's quadratic (peaks around 5, then declines). This is because initial human presence removes wolves (a key predator), but continued human presence has a major negative impact on deer. i.e., imagine a human density of ~5 corresponds to suburbs or Greenough Park, a human density of 30-40 would be more like downtown Missoula!

Here is the simulated relationship between ungulates and wolves



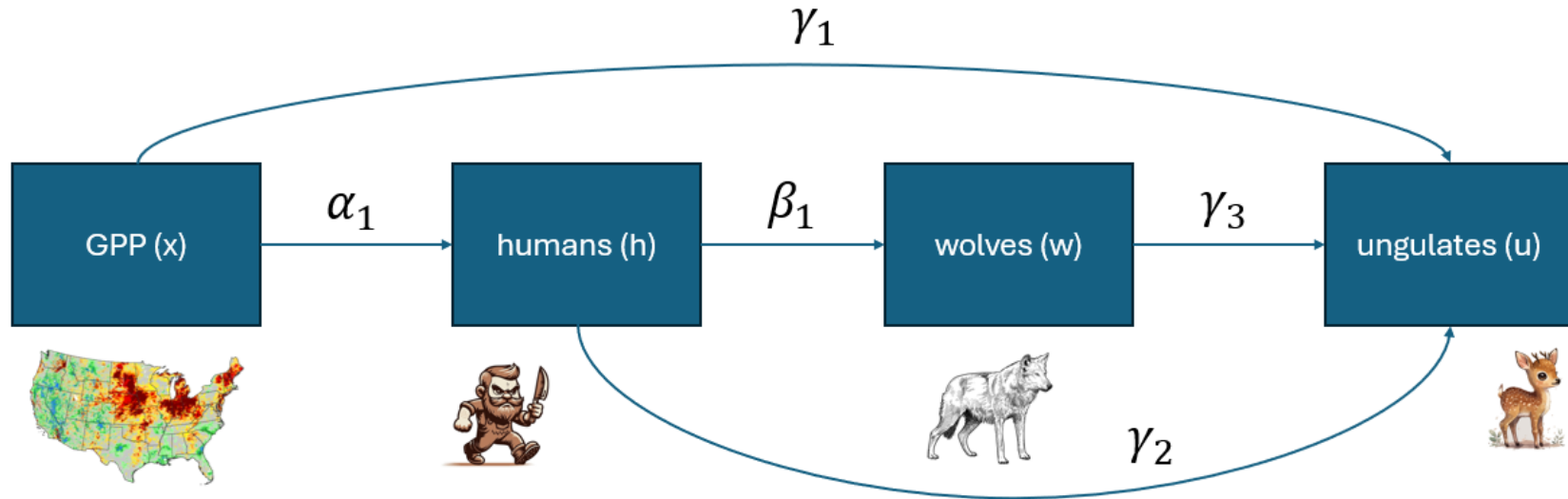
It's quadratic (hitting 0 at very low wolf densities because that's where lots of humans are) and declining at high wolf densities because wolves eat deer

Here is the simulated relationship between ungulates and GPP



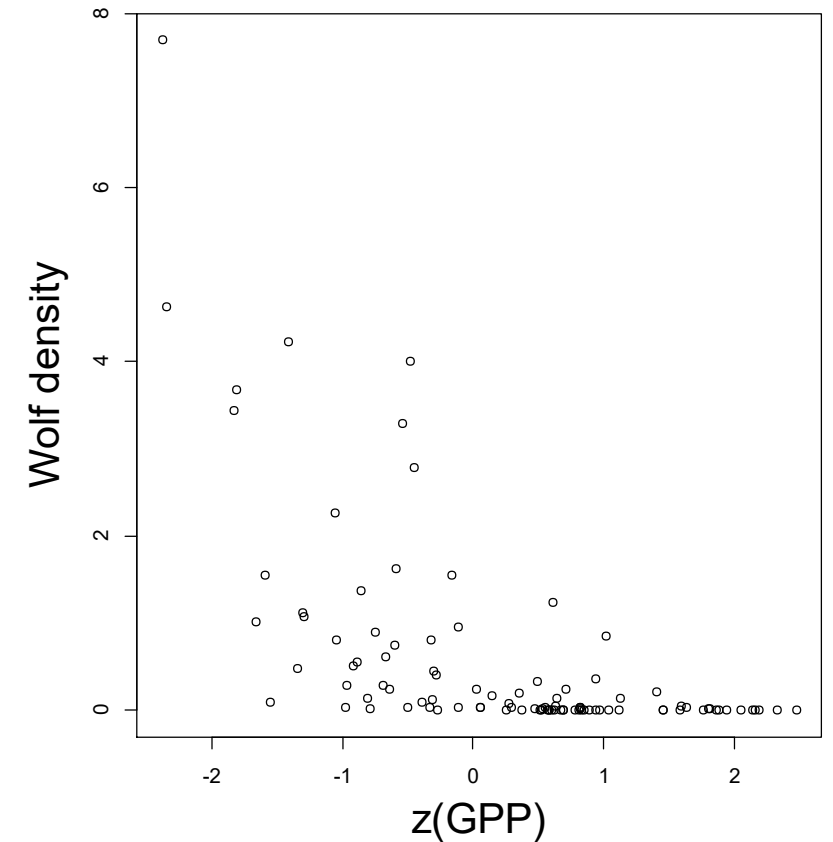
High GPP valleys have lots of people, and thus few ungulates. So, the direct effect of GPP on ungulate density (γ_1) is positive, but the 'total' effect of GPP on ungulates is negative or quadratic because high GPP leads to lots of people.

Take-home 1: complex paths produce complex signals



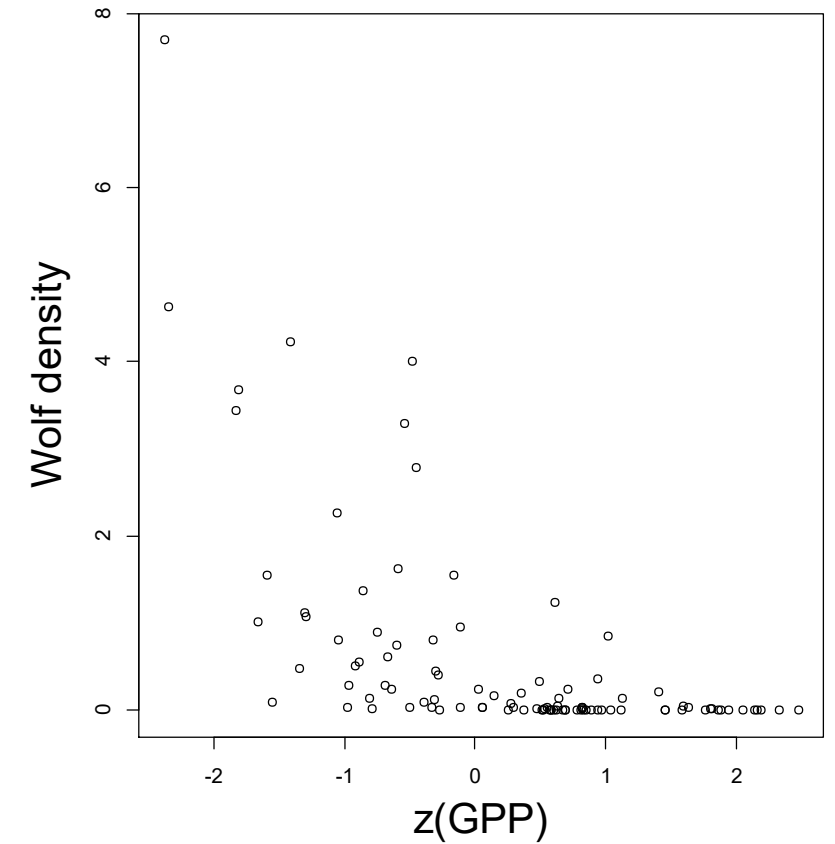
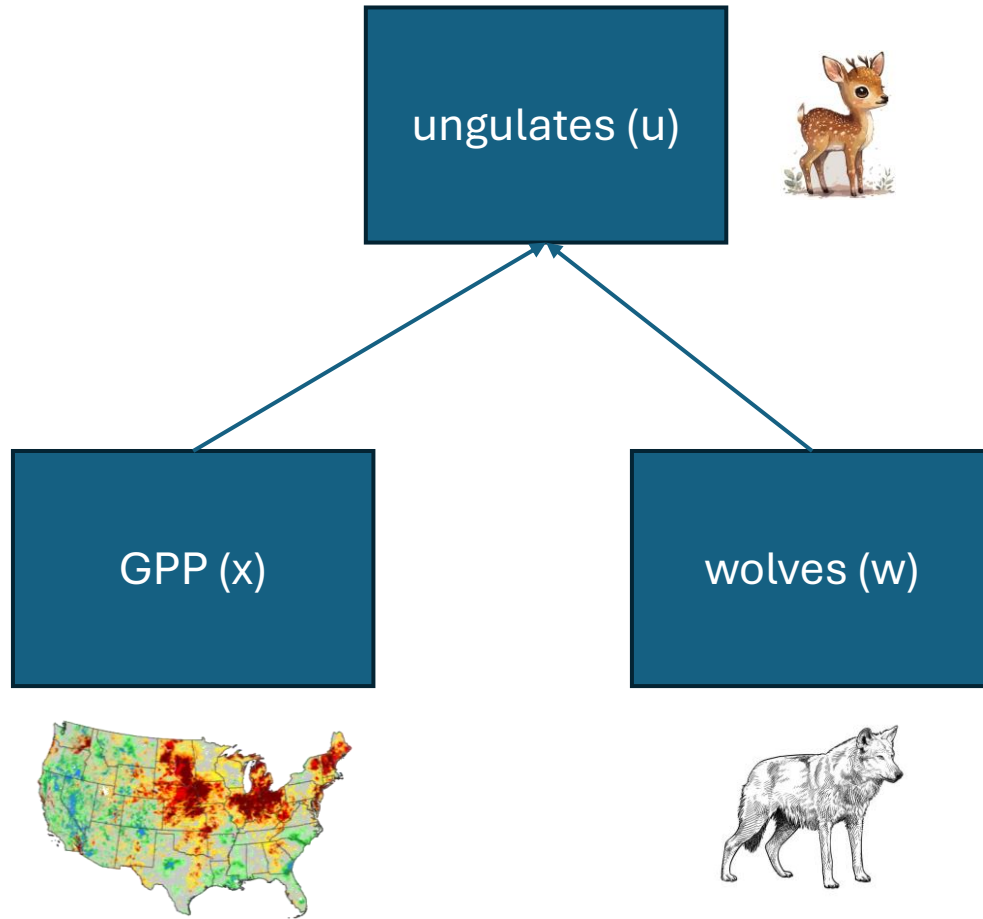
***Almost everything (worth studying) is the result of a complex path!**

Take-home 2: univariate relationships can be wildly misleading!



This can create problems for inference (it's predictive, but not causal)

e.g., imagine we wanted to build this model



This can create problems for inference